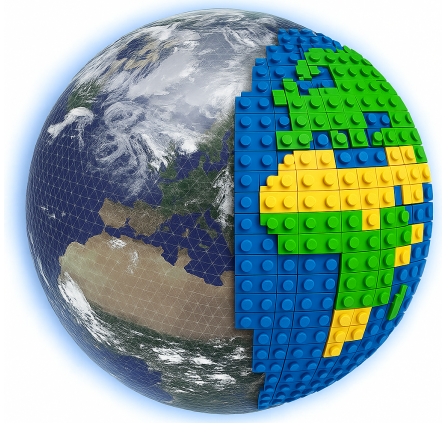




legoESM Scientific Guide

A Fully Differentiable Earth System Model in JAX

Version 5.1 — May 2026

legoESM Development Team

This document provides a comprehensive scientific description of the legoESM Earth System Model, including governing equations, numerical methods, and parameterisation schemes for all model components. The structure follows the NCAR CAM Scientific Guide convention of self-contained chapters for dynamics, physics, ocean, land, and sea ice.

Contents

1	Introduction	4
1.1	Overview	4
1.2	Design Principles	4
1.3	Notation and Conventions	5
2	Grid Systems and Coordinates	6
2.1	Cubed-Sphere Grid	6
2.1.1	Face-to-Cartesian Mapping	6
2.1.2	Cell Area	6
2.1.3	Grid Spacing	6
2.1.4	Grid Angle and Wind Rotation	6
2.1.5	Halo Exchange	7
2.2	Gaussian Grid and Spectral Transforms	7
2.2.1	Grid Dimensions	7
2.2.2	Associated Legendre Polynomials	7
2.2.3	Spherical Harmonic Transforms	7
2.2.4	Spectral Operators	8
2.2.5	Velocity from Vorticity and Divergence	8
2.3	Lat-Lon Grid	8
2.4	Voronoi/MPAS Grid	8
2.5	Vertical Coordinates	8
2.5.1	Sigma Coordinate	8
2.5.2	Hybrid Sigma-Pressure Coordinate	9
2.5.3	Terrain-Following Height Coordinate (z^*)	9
2.5.4	Ocean z^* Coordinate	9
3	Coupling of Dynamics and Physics	10
3.1	Physics Pipeline	10
3.2	Radiation Sub-Cycling	10
3.3	Grid-Agnostic Column Adapter	10
3.4	Time-Split vs Process-Split	10
4	Atmospheric Dynamics	12
4.1	Shallow Water Equations	12
4.1.1	C-D Grid Cubed-Sphere Discretisation	12
4.1.2	Lat-Lon Finite Volume Discretisation	14
4.1.3	C-Grid Lat-Lon Discretisation	14

4.1.4	TRiSK Discretisation on Voronoi Mesh	14
4.1.5	Spectral Shallow Water	14
4.2	Hydrostatic Primitive Equations	15
4.2.1	Continuous Equations	15
4.2.2	C-D Grid Cubed-Sphere Discretisation	15
4.2.3	Spectral Transform Discretisation	15
4.2.4	Lat-Lon Finite-Volume Discretisation	16
4.2.5	MPAS/Voronoi Discretisation	16
4.3	Non-Hydrostatic Compressible Euler Equations	16
4.3.1	Split-Explicit Time Integration	16
4.3.2	Sponge Layer	17
4.4	Finite-Volume Transport	17
4.4.1	MUSCL Reconstruction (2nd Order)	17
4.4.2	PPM Reconstruction (4th Order)	17
4.4.3	WENO5 Reconstruction (5th Order)	17
4.4.4	Rusanov Flux	18
4.5	Time Integration	18
4.5.1	SSP-RK3	18
4.5.2	Robert–Asselin–Williams (RAW) Time Filter	18
4.5.3	CFL Conditions and Adaptive Hyperdiffusion	18
5	Atmospheric Physics	19
5.1	Thermodynamic Preliminaries	19
5.1.1	Saturation Mixing Ratio	19
5.1.2	Virtual Temperature	19
5.1.3	Moist Adiabatic Lapse Rate	19
5.2	Radiation	19
5.2.1	Gray Two-Stream Radiation	19
5.2.2	RRTMGP Correlated- k Radiation	20
5.2.3	Solar Geometry	20
5.3	Deep Convection	20
5.3.1	Simplified Betts–Miller (SBM) Relaxation	21
5.3.2	Kuo Column Moisture-Excess Scheme	21
5.3.3	Mass-Flux Convection (Arakawa–Wu Type)	21
5.3.4	EDMF Convection	22
5.3.5	Deep Convective Adjustment	22
5.4	Cloud Microphysics	22
5.4.1	Kessler Warm-Rain Scheme	22
5.4.2	Seifert–Beheng Two-Moment Warm Rain	23
5.4.3	Morrison Double-Moment (Ice + Liquid)	23
5.4.4	Thompson Hybrid-Moment Microphysics	24
5.4.5	Summary of Microphysics Schemes	24
5.5	Cloud Fraction	24
5.5.1	Sundqvist (1988)	24
5.5.2	Xu–Randall (1996)	24
5.6	Boundary-Layer Turbulence	25
5.6.1	Louis (1979) Local Closure	25
5.6.2	Holtslag–Boville Nonlocal K-Profile	25

5.6.3	TKE-Based Closure	25
5.6.4	Implicit Vertical Diffusion	26
5.7	Surface Exchange Formulations	26
5.7.1	Neutral Bulk Aerodynamic Fluxes	26
5.7.2	Monin–Obukhov Similarity Theory (MOST)	26
5.7.3	COARE 3.0 Roughness	26
5.7.4	Large & Yeager (2004) Roughness	27
5.8	Gravity Wave Drag	27
5.8.1	Orographic GWD (Lindzen 1981)	27
5.9	Held–Suarez Forcing	27
5.10	Additional Turbulence Schemes	27
5.10.1	Smagorinsky	27
5.10.2	YSU (Yonsei University)	28
5.10.3	CLUBB-lite	28
5.10.4	EDMF Turbulence	28
5.11	Additional Convection Schemes	28
5.11.1	Kuo (1974)	28
5.11.2	Deep Convective Adjustment (DCA)	28
5.12	Gravity Wave Drag Schemes	28
5.13	Neural and Learned Physics	28
5.13.1	NeuralPhysics	28
5.13.2	LearnedColumn	29
6	Ocean Model	30
6.1	Boussinesq Hydrostatic Primitive Equations	30
6.1.1	Tracer Equations (Skew-Symmetric)	30
6.1.2	Free-Surface Equation	30
6.2	Barotropic Solver	30
6.3	Equation of State (Wright 1997)	31
6.4	Hydrostatic Pressure and Vertical Velocity	31
6.5	Vertical Mixing: K-Profile Parameterisation (KPP)	31
6.5.1	Boundary Layer Depth	31
6.5.2	K-Profile Shape	31
6.5.3	Non-Local Transport	32
6.6	Lateral Mixing	32
6.6.1	Harmonic (Laplacian) Diffusion	32
6.6.2	Biharmonic (∇^4) Diffusion	32
6.6.3	Gent–McWilliams / Redi Parameterisation	32
6.7	Tracer Advection Schemes	32
6.8	Slab and Two-Layer Ocean	34
6.9	Realistic-Geometry Spinup on ETOPO Bathymetry	34
7	Ocean Biogeochemistry	36
7.1	Carbonate Chemistry and Air–Sea CO ₂ Exchange	36
7.1.1	Gas Transfer Velocity	36
7.1.2	Net Air–Sea Flux	36
7.2	NPZD Ecosystem Model	36
7.3	Ocean Bottom Drag	37

7.4	Ocean Surface Forcing	37
7.5	Shortwave Penetration	37
7.6	Ocean Convection	37
7.6.1	Enhanced Diffusion	37
7.6.2	Plume Model	37
7.7	Richardson-Number Vertical Mixing	38
8	Land Surface Model	39
8.1	Slab Land	39
8.1.1	Surface Energy Balance	39
8.1.2	Bucket Hydrology	39
8.2	Multi-Layer Land Model	39
8.2.1	Soil Thermal Diffusion	39
8.2.2	Richards Equation (Soil Hydraulics)	39
8.2.3	Snow Budget	40
8.3	Land Carbon Cycle	40
8.3.1	DALEC-990 Six-Pool Model	40
8.3.2	Farquhar (1980) Photosynthesis	40
8.3.3	Stomatal Conductance	41
8.4	Plant Functional Types	41
8.4.1	Parameter Lookup	41
8.4.2	Trainable Parameters	41
8.4.3	Neural Parameter Provider	41
8.5	DifferLand: Differentiable Land Model Training	42
8.5.1	Phenology	42
8.5.2	Decomposition	42
8.5.3	NPP Allocation	42
9	Sea Ice	43
9.1	Thermodynamic Slab	43
9.1.1	Surface Temperature	43
9.1.2	Growth and Melt	43
9.1.3	Concentration	43
9.2	EVP Rheology	43
9.3	Multi-Category Ice Thickness Distribution	43
9.4	Free-Drift Dynamics	44
9.5	Ice Tracer Transport	44
10	Tile-Based Coupler	45
10.1	Surface Flux Blending	45
10.2	Ocean Albedo	45
10.3	Lake Model	45
10.3.1	Energy Balance	45
10.3.2	Freezing	45

11 Machine Learning Components	46
11.1 Spherical Fourier Neural Operator (SFNO)	46
11.1.1 Architecture	46
11.1.2 Spectral Convolution	46
12 Data Assimilation	47
12.1 Incremental 4D-Var	47
12.1.1 Control Vector	47
12.1.2 Cost Function	47
12.1.3 Background Error Covariance	47
12.1.4 Minimisation	48
12.1.5 Observation Operators	48
12.1.6 Analysis–Forecast Cycling	48
12.1.7 Incremental Analysis Update	48
13 Conservation and Numerical Corrections	49
13.1 Atmospheric Mass Conservation	49
13.2 Ocean Conservation	49
14 Differentiable Training	50
14.1 Physics Parameter Tuning	50
14.2 Neural GCM Training	50
14.3 SFNO–Dycore Coupling	50
14.4 ERA5 Data Ingestion	50
15 Parallel Runtime	51
15.1 Execution Modes	51
15.2 Domain Decomposition	51
15.3 MPI Differentiability	51
15.4 Ensemble Parallelism	52
15.5 Multi-Device Collective Reductions	52
16 CMIP Infrastructure	53
16.1 CF/CMOR-Compliant Output	53
16.2 Experiment Templates	53
16.3 AMIP CMIP6 Forcing Deck	53
16.4 Restart and Reproducibility	55
17 Verification and Testing	56
17.1 Williamson Standard Test Cases	56
17.2 Spectral Dynamical Core Test Suite	56
17.3 Cross-Discretisation Intercomparison	56
17.4 MPI Distributed Tests	57
18 Physical Constants	58

Chapter 1

Introduction

1.1 Overview

legoESM is a modular, fully differentiable Earth System Model (ESM) written entirely in JAX. Every component—atmosphere, ocean, land, sea ice, lake, and biogeochemistry—is compatible with JAX’s automatic differentiation (`jit`, `grad`, `vmap`, `scan`), enabling novel applications in data assimilation (4D-Var with exact adjoints), parameter estimation, hybrid AI–physics modelling, and sensitivity analysis.

The model features:

- **25+ dynamical cores** spanning five discretisation families (C-D grid FV3, spectral, finite-volume lat-lon, C-grid lat-lon, and MPAS/Voronoi), with shallow water, hydrostatic primitive equation, and non-hydrostatic compressible Euler equation sets;
- **35+ physics parameterisations** for radiation, convection, microphysics, boundary-layer turbulence, clouds, gravity wave drag, and surface exchange;
- A **3D Boussinesq ocean** with split-explicit barotropic solver, KPP vertical mixing, GM-Redi eddy parameterisation, and biogeochemistry (abiotic carbon cycle + NPZD ecosystem);
- A **multi-layer land model** with Richards-equation soil hydrology, Johansen thermal diffusion, Farquhar–Ball–Berry photosynthesis, and a DALEC-990 carbon cycle;
- **Sea-ice dynamics** with EVP rheology, multi-category thickness distribution, and thermodynamic coupling;
- A **tile-based coupler** with Monin–Obukhov surface layer, area-weighted flux blending, and radiation sub-cycling;
- **CMIP6-class infrastructure**: CF/CMOR output, six experiment templates, checkpointing with SHA-256 reproducibility verification, and a 16-parameter tuning registry;
- A **Spherical Fourier Neural Operator** (SFNO) for learned dynamics and physics emulation.

1.2 Design Principles

1. **JAX-first**: All code uses pure-functional JAX with no Python loops in hot paths. Conditional branching uses `jnp.where` or `jax.lax.cond` to preserve differentiability.
2. **Grid-agnostic physics**: Operators implement a shared protocol (`GridProtocol`); physics schemes accept any grid via column adapters that reshape to `(ncol, nlev)`.
3. **Modular composition**: Each parameterisation follows a factory pattern returning a callable `physics_fn(state, grid, coord) → Tendencies`.

4. **Unified C-D grid:** Atmosphere and ocean share the same FV3-style C-D grid operator module on the cubed sphere, eliminating code duplication and the Hollingsworth–Kallberg instability.
5. **Pytree registration:** Field and state objects are registered as JAX pytrees for seamless autodiff through the entire model.

1.3 Notation and Conventions

Throughout this document:

- Vectors are denoted in bold: $\mathbf{v} = (u, v)$;
- $\hat{f}$ denotes spectral (spherical harmonic) coefficients;
- Subscript k indexes vertical levels (increasing downward);
- Half-integer subscripts ($k \pm \frac{1}{2}$) denote layer interfaces;
- SI units are used unless otherwise noted;
- $f = 2\Omega \sin \phi$ is the Coriolis parameter.

Chapter 2

Grid Systems and Coordinates

2.1 Cubed-Sphere Grid

Six faces of a cube are mapped to the sphere via **gnomonic (central) projection**. Let $(\alpha_x, \alpha_y) \in [-\pi/4, \pi/4]^2$ be gnomonic coordinates on a face.

2.1.1 Face-to-Cartesian Mapping

For face 0 (+x face):

$$(x, y, z) = (1, \tan \alpha_x, \tan \alpha_y). \quad (2.1)$$

Points are normalised to the unit sphere: $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$, with geographic coordinates $\lambda = \text{atan2}(\hat{y}, \hat{x})$, $\phi = \arcsin(\hat{z})$.

2.1.2 Cell Area

$$\Delta A = \frac{R^2 \Delta \alpha^2}{\cos^2 \alpha_x \cos^2 \alpha_y D^3}, \quad D = \sqrt{1 + \tan^2 \alpha_x + \tan^2 \alpha_y}, \quad \Delta \alpha = \frac{\pi}{2n}. \quad (2.2)$$

2.1.3 Grid Spacing

Great-circle distances use the chord-to-arc formula:

$$d = R \cdot 2 \arcsin\left(\frac{|\mathbf{r}_1 - \mathbf{r}_2|}{2}\right). \quad (2.3)$$

The minimum grid spacing on a C_n cubed sphere is

$$\Delta x_{\min} = \frac{\pi R}{2n\sqrt{3}}, \quad (2.4)$$

occurring at the cube corners where the gnomonic distortion is maximal.

2.1.4 Grid Angle and Wind Rotation

The angle θ between the grid x -axis and geographic east:

$$\theta = \text{atan2}\left(\frac{\partial \phi}{\partial i}, \frac{\partial \lambda}{\partial i} \cos \phi\right). \quad (2.5)$$

Wind rotation between geographic (u_E, v_N) and grid-aligned (u_g, v_g) :

$$\begin{pmatrix} u_E \\ v_N \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} u_g \\ v_g \end{pmatrix}. \quad (2.6)$$

2.1.5 Halo Exchange

Scalar fields are exchanged directly across the six-face connectivity table. Vector fields are first rotated to geographic components, padded as scalars, then rotated back. Two halo widths are supported: `halo=1` for second-order and MUSCL stencils, and `halo=2` for PPM/WENO5 and biharmonic operators.

Native 4D halo exchange. All 3D operators use a 4D halo path (`pad_halo_4d`, `pad_halo_vector_4d`) that exchanges all vertical levels in a single MPI message per neighbour, replacing the earlier `vmap(pad_halo)` path which issued n_{lev} separate messages.

MPI differentiability. All `sendrecv` calls go through a `@jax.custom_vjp` wrapper (`_sendrecv_vjp` in `halo_exchange.py`) that swaps source and destination ranks in the backward pass, enabling `jax.grad` through distributed halo exchange. Only `allreduce(SUM)` is AD-safe among MPI reductions; `MAX/MIN/allgather/bcast` are used only in diagnostics and are not differentiable.

2.2 Gaussian Grid and Spectral Transforms

The spectral dynamical cores use a Gaussian grid with triangular truncation at total wavenumber n_{max} .

2.2.1 Grid Dimensions

$$n_{\text{lat}} = \lceil \frac{3}{2}(n_{\text{max}} + 1) \rceil_{\text{even}}, \quad n_{\text{lon}} = 2 n_{\text{lat}}, \quad n_{\text{sh}} = \frac{(n_{\text{max}} + 1)(n_{\text{max}} + 2)}{2}. \quad (2.7)$$

Latitudes are Gaussian quadrature points $\mu_k = \sin \phi_k$ with weights w_k ; longitudes are equispaced.

2.2.2 Associated Legendre Polynomials

Fully normalised (4π) polynomials $P_n^m(\mu)$ satisfy $\int |Y_n^m|^2 dA = 1$. Computed via the recursion:

$$P_m^m = -\sqrt{\frac{2m+1}{2m}} \sqrt{1-\mu^2} P_{m-1}^{m-1}, \quad (2.8)$$

$$P_n^m = \sqrt{\frac{4n^2-1}{n^2-m^2}} \left(\mu P_{n-1}^m - \sqrt{\frac{(n-1)^2-m^2}{4(n-1)^2-1}} P_{n-2}^m \right). \quad (2.9)$$

The derivative operator $H_n^m = -(1-\mu^2) dP_n^m/d\mu$ is used for meridional gradients.

2.2.3 Spherical Harmonic Transforms

Analysis (grid $\rightarrow$ spectral).

$$\hat{f}_{nm} = 2\pi \sum_k w_k P_n^m(\mu_k) \tilde{f}_m(\mu_k), \quad \tilde{f}_m = \frac{1}{n_{\text{lon}}} \text{FFT}[f]_m. \quad (2.10)$$

Synthesis (spectral $\rightarrow$ grid).

$$f(\phi, \lambda) = \text{iFFT} \left[\sum_{n=m}^{n_{\max}} \hat{f}_{nm} P_n^m(\mu) \right]. \quad (2.11)$$

2.2.4 Spectral Operators

Laplacian eigenvalue: $\nabla^2 \hat{f}_{nm} = -n(n+1)/a^2 \cdot \hat{f}_{nm}$. Inverse Laplacian: $\nabla^{-2} \hat{f}_{nm} = -a^2/[n(n+1)]$ for $n > 0$. Hyperdiffusion: $-\nu[n(n+1)/a^2]^p \hat{f}_{nm}$.

2.2.5 Velocity from Vorticity and Divergence

Streamfunction $\psi = \nabla^{-2} \zeta$, velocity potential $\chi = \nabla^{-2} \delta$. Velocity reconstruction:

$$u \cos \phi = \frac{\partial \psi}{\partial \theta} + \frac{\partial \chi}{\partial \lambda}, \quad v \cos \phi = \frac{\partial \psi}{\partial \lambda} - \frac{\partial \chi}{\partial \theta}, \quad (2.12)$$

where θ is colatitude.

2.3 Lat-Lon Grid

A uniform $n_{\text{lat}} \times n_{\text{lon}}$ latitude-longitude grid with A-grid (collocated) staggering. All operators use second-order centred differences. A Fourier polar filter is applied to tendencies at high latitudes to prevent the CFL constraint from becoming prohibitively small near the poles.

2.4 Voronoi/MPAS Grid

An unstructured Voronoi tessellation generated from an icosahedral parent mesh. The TRiSK (Thuburn–Ringle–Skamarock–Klemp) scheme provides mimetic discrete operators on this grid:

- Scalars (height, tracers, pressure) at cell centres;
- Normal velocity at edges;
- Vorticity at vertices (triangle dual).

Variable resolution is supported by mesh refinement.

2.5 Vertical Coordinates

2.5.1 Sigma Coordinate

$$\sigma = p / p_s, \quad \sigma \in [\sigma_{\text{top}}, 1]. \quad (2.13)$$

Half-level interfaces $\sigma_{k+1/2}$ are uniformly spaced; full levels $\sigma_k = (\sigma_{k-1/2} + \sigma_{k+1/2})/2$.

Simmons–Burridge Geopotential

The geopotential at full level k is

$$\Phi_k = \Phi_s + R_d \sum_{k'=k+1}^{N-1} T_{k'} \ln \frac{\sigma_{k'+1/2}}{\sigma_{k'-1/2}} + \alpha_k R_d T_k, \quad (2.14)$$

where the Simmons–Burridge coefficient is

$$\alpha_k = 1 - \frac{\sigma_{k-1/2}}{\Delta\sigma_k} \ln \frac{\sigma_{k+1/2}}{\sigma_{k-1/2}}. \quad (2.15)$$

Sigma-Dot (Vertical Velocity)

$$\dot{\sigma}_{k+1/2} = \frac{\sigma_{k+1/2} - \sigma_{\text{top}}}{1 - \sigma_{\text{top}}} D_{\text{tot}} - \sum_{k'=0}^k \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}, \quad D_{\text{tot}} = \sum_{k'=0}^{N-1} \nabla \cdot \mathbf{v}_{k'} \Delta\sigma_{k'}. \quad (2.16)$$

Pressure Velocity

$$\omega \equiv \frac{dp}{dt} = \sigma \frac{dp_s}{dt} + p_s \dot{\sigma}. \quad (2.17)$$

2.5.2 Hybrid Sigma-Pressure Coordinate

$$p(k) = A(k) p_{\text{ref}} + B(k) p_s, \quad (2.18)$$

where $A(k) \rightarrow 1$, $B(k) \rightarrow 0$ at the model top (pure pressure levels) and $A(k) \rightarrow 0$, $B(k) \rightarrow 1$ near the surface (pure sigma). This eliminates large pressure-gradient force errors over steep topography that afflict the pure sigma coordinate.

2.5.3 Terrain-Following Height Coordinate (z^*)

Used by the non-hydrostatic compressible Euler equations:

$$z^* = H \frac{z - z_s}{H - z_s}, \quad J = \frac{\partial z}{\partial z^*} = \frac{H - z_s}{H}. \quad (2.19)$$

A 1D reference state $(\rho_0(z), \theta_0(z), \pi_0(z))$ is pre-computed for the perturbation formulation.

2.5.4 Ocean z^* Coordinate

The ocean uses a dynamic z^* coordinate with free-surface elevation η :

$$J(\eta) = \frac{\eta + H}{H_{\text{max}}}, \quad (2.20)$$

where H is the resting depth.

Chapter 3

Coupling of Dynamics and Physics

3.1 Physics Pipeline

The `PhysicsPipeline` class composes all physical parameterisations into a single callable that is invoked once per dynamics timestep. The calling sequence within a single step is:

1. **Dynamics step:** Update winds, temperature, surface pressure, and tracers via the selected dynamical core.
2. **Physics step:** Apply parameterisations in the following order:
 - (a) Radiation (with optional sub-cycling),
 - (b) Deep convection,
 - (c) Shallow convection / large-scale condensation,
 - (d) Cloud microphysics,
 - (e) Boundary-layer turbulence and vertical diffusion,
 - (f) Gravity wave drag,
 - (g) Surface fluxes (tile coupler).
3. **Conservation fixers:** Mass, moisture, and optionally energy.
4. **Diagnostics:** Energy budget, monthly means, snapshots.

3.2 Radiation Sub-Cycling

Radiation is the most expensive physics component. The pipeline supports calling radiation every N_{rad} dynamics steps while holding radiative heating rates constant between calls. The branching is implemented via `jax.lax.cond` with both branches returning identical array shapes and dtypes.

3.3 Grid-Agnostic Column Adapter

The `ColumnAdapter` reshapes any grid’s 3D state into a `(ncol, nlev)` column format for physics, then unflattens back to the native grid shape. This allows all physics schemes to operate on simple column arrays regardless of the underlying horizontal discretisation.

3.4 Time-Split vs Process-Split

legoESM uses **time-split** coupling: dynamics and physics operate sequentially within each timestep. Physics tendencies are computed from the dynamics-updated state and applied as a forward-Euler

update. This is natural for the finite-volume dycores that use a Lagrangian vertical coordinate internally.

Chapter 4

Atmospheric Dynamics

This chapter describes the governing equations and numerical methods for each dynamical core. Cores are organised by equation system (shallow water, hydrostatic primitive equations, non-hydrostatic compressible Euler), then by discretisation.

4.1 Shallow Water Equations

The rotating shallow water equations in **vector-invariant form**:

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h \mathbf{v}), \quad (4.1)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} + D_u, \quad (4.2)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} + D_v, \quad (4.3)$$

where $B = \frac{1}{2}(u^2 + v^2) + g(h + h_s)$ is the Bernoulli function, $\zeta = \partial v / \partial x - \partial u / \partial y$ is the relative vorticity, and D_u, D_v represent hyperdiffusion.

4.1.1 C-D Grid Cubed-Sphere Discretisation

This is the primary shallow water solver, using the FV3-style C-D grid staggering of [Lin \(2004\)](#) and [Putman and Lin \(2007\)](#).

Prognostic variables. D-grid winds (u_d, v_d) at cell corners, shape $(6, n+1, n+1)$; height h at cell centres, shape $(6, n, n)$.

D-grid vorticity. Vorticity at each cell corner is the discrete circulation divided by the corner dual-cell area:

$$\zeta_{\text{corner}} = \frac{1}{A_c} \oint \mathbf{u} \cdot d\mathbf{l} \approx \frac{1}{A_c} [u_{i,j}^D \Delta x - u_{i+1,j}^D \Delta x + v_{i,j+1}^D \Delta y - v_{i,j}^D \Delta y]. \quad (4.4)$$

This circulation-based discretisation is exact on the D-grid and **eliminates the Hollingsworth–Kallberg instability** that plagues A-grid vorticity operators.

Arakawa–Lamb gradient. The Bernoulli gradient at D-grid corners uses a 4-point average:

$$\left. \frac{\partial B}{\partial x} \right|_{i+\frac{1}{2},j+\frac{1}{2}} = \frac{B_{i+1,j} + B_{i+1,j+1} - B_{i,j} - B_{i,j+1}}{2\Delta x}. \quad (4.5)$$

C-grid mass flux. C-grid velocities are diagnosed from D-grid winds by face-edge averaging. Scalar transport uses upwind C-grid mass fluxes:

$$\frac{dh}{dt} = -\frac{1}{A} [(hu_c)_{i+\frac{1}{2},j} - (hu_c)_{i-\frac{1}{2},j} + (hv_c)_{i,j+\frac{1}{2}} - (hv_c)_{i,j-\frac{1}{2}}]. \quad (4.6)$$

FV3-faithful PPM transport (v3.9). As of v3.9, the C-D grid operator module underwent a complete rewrite to implement FV3-faithful Piecewise Parabolic Method (PPM) transport (Colella and Woodward, 1984) on staggered D/C grids. Key elements include:

- PPM reconstruction with Colella–Woodward monotonicity limiters for all scalar fields;
- Flux-form mass transport on staggered D-grid and C-grid;
- Corrected geographic-to-grid wind rotation eliminating sign errors in cross-face interpolation;
- Tuned hyperdiffusion with physical e -folding time ($\nu_4 = \Delta x^4 / \tau_{\text{efold}}$).

The module spans approximately 740 lines of JAX code.

FV3 forward–backward shallow-water core (v4.0). An FV3-exact implementation of the complete `c_sw/d_sw` forward–backward stepping algorithm is provided in `fv3_sw_core.py`. The forward step (`c_sw`) computes C-grid mass fluxes and height tendencies; the backward step (`d_sw`) updates D-grid winds using the vorticity flux and Bernoulli gradient.

d2a2c_vect. The D-grid $\rightarrow$ A-grid $\rightarrow$ C-grid vector conversion faithfully ports the GFDL FV3 algorithm, including:

- Edge-boundary stencils with one-sided 4-point interpolation at cube-face edges;
- Non-orthogonality corrections via $\cos \alpha$ and $(\sin^2 \alpha)^{-1}$ corner metrics;
- Halo-aware padding with face-boundary detection.

PPM wind and vorticity transport. C-grid wind tendencies (`xtp_u`, `ytp_v`) and absolute vorticity flux use Piecewise Parabolic Method reconstruction with FV3-specific interpolation coefficients (A_1 , A_2 , C_1 – C_3) and Colella–Woodward monotonicity limiters.

Cell-centred kinetic energy. Kinetic energy is computed from A-grid velocities at cell centres, with an optional 4th-order averaging stencil (`a2b_ord4`).

Duo-grid sub-grid metrics (Mouallem et al. 2023). The cubed-sphere C-D grid computes 9-point sub-grid angle metrics per cell:

$$\sin \alpha_{\text{sg}}, \cos \alpha_{\text{sg}} \in \mathbb{R}^{6 \times n \times n \times 9}, \quad (4.7)$$

encoding the local deviation from orthogonality at cell centres, edge midpoints, and corners. These enable FV3-faithful flux scaling at cube-face boundaries where the gnomonic projection introduces non-orthogonality. Additional C-D grid metrics:

- Centre-to-centre distances Δx_c (shape $6 \times (n+1) \times n$) and Δy_c (shape $6 \times n \times (n+1)$);
- Non-orthogonality factors $\cos \alpha_{\text{corner}}$ and $(\sin^2 \alpha_{\text{corner}})^{-1}$ at D-grid corners.

Fidelity calibration and 3D edge-artifact sentinels. The C-D-grid FV3 port is gated on a dual-target Williamson 2 / Williamson 5 calibration: at C36 with $\Delta t = 300$ s the production knobs `div_damp_order = 8` and `damp_v = 0.030` (iter-1030) satisfy both criteria simultaneously — Williamson 2 v-wind error $|v_{\parallel}| < 0.119 \text{ m s}^{-1}$ and Williamson 5 day-5 height $h > 0$ with $|u| < 80 \text{ m s}^{-1}$. A 48-test sentinel suite exercises the full 3D coverage matrix (hydrostatic / non-hydrostatic $\times$ rest / edge / long integration $\times$ cube-edge $i, j \in \{0, n-1\}$) and a long 100-step cosine-bell closure within $\pm 5\%$. Visual verification of v -wind snapshots is mandatory — error norms can improve while edge artifacts shift in character; see `docs/fv3_fortran_fidelity_review.md` and `docs/cubed_sphere_edge_artifacts.md` for the iter-1045 status snapshot.

4.1.2 Lat-Lon Finite Volume Discretisation

A-grid staggering on a uniform $n_{\text{lat}} \times n_{\text{lon}}$ grid. Centred differences for all operators. A Fourier polar filter damps meridional wavelengths near the poles to maintain CFL stability.

4.1.3 C-Grid Lat-Lon Discretisation

Height at cell centres; u at longitude interfaces; v at latitude interfaces. Advantages: exact pressure gradient (just cell differences), mass flux naturally at cell interfaces (no interpolation), and better gravity wave dispersion. The Coriolis term requires 4-point interpolation.

4.1.4 TRiSK Discretisation on Voronoi Mesh

Following [Ringler et al. \(2010\)](#), the discrete operators on the Voronoi mesh are:

$$\text{Divergence at cell } c : (\nabla \cdot \mathbf{u})_c = \frac{1}{A_c} \sum_e n_{e,c} u_e l_e, \quad (4.8)$$

$$\text{Gradient at edge } e : (\nabla \phi)_e = \frac{\phi_{c2} - \phi_{c1}}{d_e}, \quad (4.9)$$

$$\text{Curl at vertex } v : (\nabla \times \mathbf{u})_v = \frac{1}{A_v} \sum_e t_{e,v} u_e d_e, \quad (4.10)$$

where $n_{e,c}$ and $t_{e,v}$ are orientation signs, l_e is edge length, d_e is distance between adjacent cell centres, and A_c, A_v are cell/vertex areas.

4.1.5 Spectral Shallow Water

Vorticity–divergence formulation on the Gaussian grid:

$$\frac{\partial \hat{\zeta}}{\partial t} = -\nabla \cdot [\widehat{(\zeta + f)} \mathbf{v}], \quad (4.11)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \times [\widehat{(\zeta + f)} \mathbf{v}] - \nabla^2 (E + \widehat{\Phi} + \Phi_s), \quad (4.12)$$

$$\frac{\partial \hat{\Phi}}{\partial t} = -\nabla \cdot (\widehat{\Phi} \mathbf{v}), \quad (4.13)$$

where $E = \frac{1}{2}(u^2 + v^2)$ and $\Phi = gh$. Pseudospectral: nonlinear products computed on the grid, then transformed to spectral space.

4.2 Hydrostatic Primitive Equations

Prognostic variables: u, v (horizontal winds), T (temperature), p_s (surface pressure), and tracers q_i .

4.2.1 Continuous Equations

In σ -coordinates:

$$\frac{\partial p_s}{\partial t} = - \int_0^1 \nabla \cdot (p_s \mathbf{v}) d\sigma, \quad (4.14)$$

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial B}{\partial x} - R_d T \frac{\partial \ln p_s}{\partial x} + D_u, \quad (4.15)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial B}{\partial y} - R_d T \frac{\partial \ln p_s}{\partial y} + D_v, \quad (4.16)$$

$$\frac{\partial T}{\partial t} = -\nabla \cdot (T \mathbf{v}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \frac{\partial T}{\partial \sigma} + \kappa T \frac{\omega}{p} + Q, \quad (4.17)$$

where $B = K + \Phi$, $K = \frac{1}{2}(u^2 + v^2)$, and $\kappa = R_d/c_p$. The adiabatic heating term $\kappa T \omega/p$ represents compression warming during descent.

4.2.2 C-D Grid Cubed-Sphere Discretisation

Prognostic state: u_d, v_d at D-grid corners $(6, n+1, n+1, N_{\text{lev}})$; T and p_s at cell centres. Spatial operators are identical to the shallow water C-D grid (Section 4.1), extended to 3D with:

- Simmons–Burridge geopotential integration (Section 2.5);
- Sigma-dot computation with discrete continuity closure;
- Upwind vertical advection: $-\dot{\sigma} \partial f / \partial \sigma$.

4.2.3 Spectral Transform Discretisation

On the Gaussian grid, the momentum equations are recast in the vorticity–divergence form (Bourke, 1972):

$$\frac{\partial \hat{\zeta}}{\partial t} = \nabla \times [\widehat{(\zeta + f)} \mathbf{v}] + \widehat{\text{vert. adv.}} - \nu \nabla^2 \hat{\zeta}, \quad (4.18)$$

$$\frac{\partial \hat{\delta}}{\partial t} = \nabla \cdot [\widehat{(\zeta + f)} \mathbf{v}] - \nabla^2 (K + \Phi + R_d T \ln p_s) + \widehat{\text{vert. adv.}} - \nu \nabla^2 \hat{\delta}, \quad (4.19)$$

$$\frac{\partial \hat{T}}{\partial t} = -\nabla \cdot (\widehat{T \mathbf{v}}) + T \nabla \cdot \mathbf{v} - \dot{\sigma} \widehat{\partial T / \partial \sigma} + \widehat{\kappa T \omega / p}, \quad (4.20)$$

$$\frac{\partial \ln p_s}{\partial t} = - \int_0^1 \hat{\delta} d\sigma. \quad (4.21)$$

Spectral curl and divergence. For flux vectors $(U, V) = (F_u \cos \phi, F_v \cos \phi)$:

$$\widehat{\text{curl}} = \frac{im}{a} \mathcal{S}[V / \cos^2 \phi] + \frac{1}{a} \mathcal{S}_\mu[U], \quad \widehat{\text{div}} = \frac{im}{a} \mathcal{S}[U / \cos^2 \phi] - \frac{1}{a} \mathcal{S}_\mu[V], \quad (4.22)$$

where $\mathcal{S}$ and $\mathcal{S}_\mu$ denote the SH analysis with standard and $dP_n^m/d\mu$ Legendre matrices. The $\cos \phi$ weighting is absorbed into the Legendre basis, making the formulation pole-safe.

Semi-implicit treatment of gravity waves. Following Hoskins and Simmons (1975), the fast gravity-wave terms are treated implicitly. Linearising about a reference temperature $T_{\text{ref}} = 300$ K yields a Helmholtz equation per spectral mode:

$$[\mathbf{I} + \alpha^2 \Delta t^2 \lambda_n \mathbf{\Gamma}] \hat{\delta}^{n+1} = \hat{\delta}^{\text{explicit}}, \quad (4.23)$$

where $\lambda_n = n(n+1)/a^2$ is the Laplacian eigenvalue and $\mathbf{\Gamma}$ is the vertical coupling matrix. This is pre-factored (LU decomposition) for each wavenumber n at initialisation, removing the gravity-wave CFL constraint and allowing 3–4 $\times$ larger timesteps.

Verification (v3.9). A comprehensive spectral dycore test suite validates conservation (mass, energy, enstrophy), long-integration stability under Held–Suarez forcing, spectral convergence with increasing truncation wavenumber, and semi-implicit scheme correctness. See Chapter 17 for details.

4.2.4 Lat-Lon Finite-Volume Discretisation

Identical equations to the C-D grid version, discretised on the A-grid lat-lon mesh with centred differences. Polar filter applied to all tendencies. Optional Rayleigh sponge layer above a configurable σ cutoff.

4.2.5 MPAS/Voronoi Discretisation

TRiSK operators (Section 4.1) extended to 3D. Uses RK4 or SSP-RK3 time stepping with energy- or enstrophy-conserving potential vorticity flux options.

4.3 Non-Hydrostatic Compressible Euler Equations

In the z^* terrain-following coordinate, the prognostic variables are perturbations from a hydrostatically balanced reference state $(\rho_0, \theta_0, \pi_0)$:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - c_p \theta \frac{\partial \pi'}{\partial x} + D_u - r u, \quad (4.24)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - c_p \theta \frac{\partial \pi'}{\partial y} + D_v - r v, \quad (4.25)$$

$$\frac{\partial w}{\partial t} = -\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} - r w, \quad (4.26)$$

$$\frac{\partial \theta'}{\partial t} = -\nabla \cdot (\theta \mathbf{v}_h) - \frac{w}{J} \frac{\partial \theta}{\partial z^*} + D_\theta, \quad (4.27)$$

$$\frac{\partial \rho'}{\partial t} = -\frac{1}{J} \left[\nabla_h \cdot (J \rho \mathbf{v}_h) + \frac{\partial(\rho w)}{\partial z^*} \right], \quad (4.28)$$

where $r(z)$ is the Rayleigh sponge coefficient and the Exner function perturbation is

$$\pi' = \pi_0 \left[\left(1 + \frac{\rho'}{\rho_0} \right) \left(1 + \frac{\theta'}{\theta_0} \right) \right]^{R_d/c_v} - \pi_0. \quad (4.29)$$

4.3.1 Split-Explicit Time Integration

Slow modes (SSP-RK3): advection, Coriolis, horizontal pressure gradient force, and diffusion.

Fast modes (forward-backward acoustic substeps): vertical sound/gravity waves. Typically $N_s = 6$ substeps per RK stage.

Forward step (vertical velocity).

$$w^{n+1} = w^n + \Delta t_s \left[-\frac{c_p \theta}{J} \frac{\partial \pi'}{\partial z^*} - g \frac{\theta'}{\theta_0} \right]. \quad (4.30)$$

Backward step (density, potential temperature).

$$\rho^{n+1} = \rho'^n - \Delta t_s \frac{\partial(\rho w^{n+1})}{\partial z^*}, \quad \theta^{n+1} = \theta'^n - \Delta t_s \frac{w^{n+1}}{J} \frac{\partial \theta}{\partial z^*}. \quad (4.31)$$

Acoustic off-centering. To damp vertically propagating acoustic/gravity wave noise (Skamarock and Klemp, 2008):

$$\rho_\beta^{n+1} = (1 + \beta) \rho'^{n+1} - \beta \rho'^n, \quad (4.32)$$

where $\beta \geq 0$ is the off-centering parameter (default 0.1).

4.3.2 Sponge Layer

$$r(z) = r_{\max} \sin^2 \left(\frac{\pi}{2} \frac{\max(0, z - z_{\text{bot}})}{z_{\text{top}} - z_{\text{bot}}} \right). \quad (4.33)$$

4.4 Finite-Volume Transport

4.4.1 MUSCL Reconstruction (2nd Order)

Piecewise-linear reconstruction with slope limiting (van Leer, 1979):

$$q_L = q_i + \frac{1}{2} \phi_i, \quad q_R = q_{i+1} - \frac{1}{2} \phi_{i+1}, \quad (4.34)$$

with minmod or MC (monotonised central) limiters.

4.4.2 PPM Reconstruction (4th Order)

The Piecewise Parabolic Method (Colella and Woodward, 1984) constructs a monotone parabola in each cell. Fourth-order edge values:

$$\hat{q}_{i+1/2} = \frac{7}{12}(q_i + q_{i+1}) - \frac{1}{12}(q_{i-1} + q_{i+2}). \quad (4.35)$$

Monotonicity constraint: $\hat{q}_{i+1/2} \leftarrow \text{median}(\hat{q}_{i+1/2}, q_i, q_{i+1})$. Colella-Woodward limiting prevents new extrema in the parabolic profile.

4.4.3 WENO5 Reconstruction (5th Order)

Three sub-stencil polynomials combined with nonlinear weights adapting to local smoothness (Jiang and Shu, 1996):

$$q_L = \sum_k \omega_k q^{(k)}, \quad \omega_k = \frac{\alpha_k}{\sum_j \alpha_j}, \quad \alpha_k = \frac{d_k}{(\varepsilon + \beta_k)^2}, \quad (4.36)$$

where d_k are ideal weights (1/10, 6/10, 3/10) and β_k are smoothness indicators.

4.4.4 Rusanov Flux

$$F = \frac{1}{2}(h_L u_L + h_R u_R) - \frac{1}{2} \alpha (h_R - h_L), \quad \alpha = \max(|u_L| + c_L, |u_R| + c_R), \quad (4.37)$$

where $c = \sqrt{g h}$ is the gravity-wave speed.

4.5 Time Integration

4.5.1 SSP-RK3

The three-stage strong-stability-preserving Runge–Kutta method (Shu and Osher, 1988):

$$\mathbf{u}^{(1)} = \mathbf{u}^n + \Delta t F(\mathbf{u}^n), \quad (4.38)$$

$$\mathbf{u}^{(2)} = \frac{3}{4}\mathbf{u}^n + \frac{1}{4}\mathbf{u}^{(1)} + \frac{1}{4}\Delta t F(\mathbf{u}^{(1)}), \quad (4.39)$$

$$\mathbf{u}^{n+1} = \frac{1}{3}\mathbf{u}^n + \frac{2}{3}\mathbf{u}^{(2)} + \frac{2}{3}\Delta t F(\mathbf{u}^{(2)}). \quad (4.40)$$

SSP-RK(4,3) and SSP-RK(5,4) variants are also available.

4.5.2 Robert–Asselin–Williams (RAW) Time Filter

The leapfrog scheme used by the semi-implicit spectral PE solver requires a time filter to damp the computational mode. The Williams (2009) modification:

$$d^n = \frac{\gamma}{2}(X^{n-1} - 2X^n + X^{n+1}), \quad (4.41)$$

$$X_{\text{filt}}^n = X^n + (1 - \alpha) d^n, \quad (4.42)$$

$$X_{\text{filt}}^{n+1} = X^{n+1} + \alpha d^n, \quad (4.43)$$

with $\alpha = 0.5$ (default). This restores second-order accuracy while preserving the damping of the computational mode.

4.5.3 CFL Conditions and Adaptive Hyperdiffusion

The CFL-limited timestep:

$$\Delta t_{\text{max}} = \frac{C_{\text{CFL}} \Delta x_{\text{min}}}{(u_{\text{max}} + c_{\text{gw}}) \sqrt{n_{\text{dim}}}}, \quad (4.44)$$

where $c_{\text{gw}} = \sqrt{g H}$ is the gravity wave speed.

The maximum stable hyperdiffusion coefficient for p -th order:

$$\nu_p = S \cdot \frac{C(p) \Delta x^{2p}}{\Delta t}, \quad C(2) = \frac{1}{2}, \quad C(4) = \frac{1}{8}, \quad C(6) = \frac{1}{48}, \quad (4.45)$$

where $S \in (0, 1]$ is a safety factor. The physical e-folding time for $2\Delta x$ waves is $\tau_{\text{efold}} = \Delta x^{2p} / \nu_p$.

Chapter 5

Atmospheric Physics

5.1 Thermodynamic Preliminaries

5.1.1 Saturation Mixing Ratio

Using the Tetens (August–Roche–Magnus) formula:

$$e_s(T) = 611.2 \exp\left(\frac{17.67 T_c}{T_c + 243.5}\right), \quad q_s = \frac{\varepsilon e_s}{p - e_s}, \quad \varepsilon = R_d/R_v \approx 0.622, \quad (5.1)$$

where $T_c = T - 273.15$ K. Over ice, the Clausius–Clapeyron form is used:

$$e_{s,i}(T) = 611.2 \exp\left[\frac{L_s}{R_v} \left(\frac{1}{T_{\text{freeze}}} - \frac{1}{T}\right)\right]. \quad (5.2)$$

5.1.2 Virtual Temperature

$$T_v = T (1 + 0.608 q_v - q_c - q_r - q_i). \quad (5.3)$$

5.1.3 Moist Adiabatic Lapse Rate

$$\Gamma_m = \frac{R_d T}{c_p p} \frac{1 + L_v q_s / (R_d T)}{1 + L_v^2 q_s / (c_p R_v T^2)}. \quad (5.4)$$

5.2 Radiation

5.2.1 Gray Two-Stream Radiation

Following [Frierson et al. \(2006\)](#):

Longwave Optical Depth

$$\tau(\sigma, \phi) = [f_l \sigma + (1 - f_l) \sigma^4] [\tau_e + (\tau_p - \tau_e) \sin^2 \phi] + \sum_k \tau_{\text{moist}} q_{v,k} \frac{\Delta p_k}{g}, \quad (5.5)$$

where the moisture optical depth per layer is $\Delta \tau_{\text{moist},k} = \tau_{\text{moist}} q_{v,k} \Delta p_k / g$, with $\tau_e = 7.2$, $\tau_p = 1.8$, $f_l = 0.2$, $\tau_{\text{moist}} = 0.0115 \text{ m}^2 \text{ kg}^{-1}$.

Two-Stream Longwave Fluxes

Layer transmittance and emissivity:

$$\mathcal{T}_k = e^{-D \Delta \tau_k}, \quad \varepsilon_k = 1 - \mathcal{T}_k, \quad D = 1.66. \quad (5.6)$$

The diffusivity factor $D \approx 5/3$ accounts for the hemispheric-mean angular integration.

Upward sweep from surface ($F_{\text{sfc}}^\uparrow = \varepsilon_s \sigma_{\text{SB}} T_s^4$):

$$F_k^\uparrow = \mathcal{T}_k F_{k+1}^\uparrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (5.7)$$

Downward sweep from TOA ($F_0^\downarrow = 0$):

$$F_k^\downarrow = \mathcal{T}_k F_{k-1}^\downarrow + \varepsilon_k \sigma_{\text{SB}} T_k^4. \quad (5.8)$$

Shortwave (Beer–Lambert)

$$F_{\text{SW}}^\downarrow(k) = \frac{S_0}{4} e^{-\tau_{\text{SW},0} \sigma_k}, \quad F_{\text{SW}}^\uparrow(k) = \alpha_s F_{\text{SW}}^\downarrow(\text{sfc}), \quad (5.9)$$

with $\tau_{\text{SW},0} = 0.22$, $S_0 = 1361 \text{ W m}^{-2}$, $\alpha_s = 0.31$. Only the downward beam is absorbed; reflected SW escapes to space.

Heating Rate

$$\left. \frac{dT}{dt} \right|_{\text{rad},k} = \frac{g}{c_p} \frac{F_{\text{net},k+1/2}^\uparrow - F_{\text{net},k-1/2}^\uparrow}{p_{k+1/2} - p_{k-1/2}}. \quad (5.10)$$

5.2.2 RRTMGP Correlated- k Radiation

The RRTMGP backend uses JAX-RRTMGP with:

- 128 LW and 112 SW g -points for correlated- k gas absorption;
- Two-stream delta-Eddington radiative transfer equation;
- Spectral line absorption for H_2O , CO_2 , CH_4 , N_2O , O_3 ;
- Cloud optics via liquid/ice water path and effective radii lookup tables;
- Optional aerosol optical depth with prescribed single-scattering albedo ($\omega_0 = 0.93$) and asymmetry factor ($g = 0.70$).

Default trace-gas concentrations: $\text{CO}_2=415 \text{ ppmv}$, $\text{CH}_4=1900 \text{ ppbv}$, $\text{N}_2\text{O}=332 \text{ ppbv}$.

5.2.3 Solar Geometry

The cosine of the solar zenith angle:

$$\mu_0 = \sin \phi \sin \delta + \cos \phi \cos \delta \cos h, \quad (5.11)$$

where δ is the solar declination and h is the hour angle. Daily-mean insolation and perpetual-equinox modes are also available.

5.3 Deep Convection

Five convection schemes are available, all fully JAX-differentiable.

5.3.1 Simplified Betts–Miller (SBM) Relaxation

Following Frierson (2007):

CAPE Computation

$$\text{CAPE} = R_d \sum_k \max(0, T_{\text{parcel},k} - T_{\text{env},k}) \frac{\Delta p_k}{p_k}. \quad (5.12)$$

Smooth Trigger

$$\mathcal{F} = \sigma(\kappa_s(\text{CAPE} - \text{CAPE}_c)), \quad \kappa_s = 0.01 \text{ (J/kg)}^{-1}. \quad (5.13)$$

Enthalpy-Conserving Relaxation

The reference profile T_{ref} is adjusted so that column-integrated moist enthalpy is conserved:

$$\sum_k [c_p(T_{\text{ref},k} - T_k) + L_v(q_{\text{ref},k} - q_{v,k})] \Delta p_k = 0. \quad (5.14)$$

Tendencies:

$$\frac{dT}{dt} = \mathcal{F} \frac{T_{\text{ref}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} \frac{q_{\text{ref}} - q_v}{\tau_c}, \quad \tau_c = 7200 \text{ s}. \quad (5.15)$$

Precipitation

$$P = -\frac{1}{g} \sum_k \frac{dq_v}{dt} \Delta p_k. \quad (5.16)$$

5.3.2 Kuo Column Moisture-Excess Scheme

Convection proportional to column-integrated moisture excess above saturation $[\text{kg m}^{-2}]$. Heating and moistening are partitioned via a parameter α_{heat} :

$$MC = \int \max(q_v - q_s, 0) \frac{dp}{g}, \quad (5.17)$$

$$\frac{dT}{dt} = \mathcal{F} \alpha_{\text{heat}} \frac{T_{\text{moist}} - T}{\tau_c}, \quad \frac{dq_v}{dt} = \mathcal{F} (1 - \alpha_{\text{heat}}) \frac{q_{s,\text{moist}} - q_v}{\tau_c}, \quad (5.18)$$

with $\alpha_{\text{heat}} = 0.75$ and $\tau_c = 3600 \text{ s}$.

5.3.3 Mass-Flux Convection (Arakawa–Wu Type)

A prognostic base mass flux M_c relaxes toward an equilibrium value:

$$M_{\text{eq}} = \sigma[(\text{CAPE} - \text{CAPE}_{\text{thresh}})/\text{scale}] \cdot M_{\text{scale}}, \quad \frac{dM_c}{dt} = \frac{M_{\text{eq}} - M_c}{\tau_{\text{adj}}}. \quad (5.19)$$

The mass-flux profile is sinusoidal: $m(p) = \sin[\pi(p_{\text{base}} - p)/\Delta p]$. The entraining updraft uses exponential dilution:

$$T_u(z) = e^{-\epsilon_0 z} T_{\text{moist}} + (1 - e^{-\epsilon_0 z}) T_{\text{env}}, \quad (5.20)$$

with entrainment rate $\epsilon_0 = 10^{-3} \text{ m}^{-1}$ and detrainment rate $\delta_0 = 10^{-3} \text{ m}^{-1}$.

Precipitation. Precipitation is diagnosed from detrainment of updraft condensate, integrated over the column using layer thickness Δz :

$$P = \sum_k \delta_0 M_k q_{\ell,k} \Delta z_k \quad [\text{kg m}^{-2} \text{s}^{-1}], \quad (5.21)$$

where $q_{\ell,k} = \max(q_{u,k} - q_{s,k}, 0)$ is the updraft condensate and $M_k = M_c m(p_k)$ is the mass flux profile $[\text{kg m}^{-2} \text{s}^{-1}]$. The vertical measure Δz is used (not $\Delta p/g$) because M already carries units of mass flux per unit area, so $\delta_0 M q_\ell$ has units $[\text{kg m}^{-3} \text{s}^{-1}]$ and multiplying by Δz [m] yields $[\text{kg m}^{-2} \text{s}^{-1}]$.

5.3.4 EDMF Convection

Eddy-Diffusivity Mass-Flux with a prognostic updraft area fraction a_u :

$$\frac{da_u}{dt} = \frac{a_{u,\text{eq}} - a_u}{\tau_a}, \quad (5.22)$$

combining mass-flux (updraft) and eddy-diffusivity (environmental mixing) contributions.

Precipitation. As in the mass-flux scheme (§5.4), precipitation is diagnosed from condensate detrainment using Δz :

$$P = \sum_k \delta_0 M_{u,k} q_{\ell,k} \Delta z_k \quad [\text{kg m}^{-2} \text{s}^{-1}], \quad (5.23)$$

with $M_{u,k} = \rho_k a_u w_{u,k}$ and $\delta_0 = 2 \times 10^{-3} \text{ m}^{-1}$.

5.3.5 Deep Convective Adjustment

The simplest parameterisation: scans bottom-to-top adjusting unstable layer pairs toward moist-adiabatic neutrality while conserving total enthalpy. Uses sigmoid activation for smooth, differentiable blending.

5.4 Cloud Microphysics

5.4.1 Kessler Warm-Rain Scheme

Prognostic variables: q_c (cloud water), q_r (rain).

Saturation adjustment. The saturation excess is converted to a tendency (rate) by dividing by the time step Δt :

$$\left. \frac{dq_v}{dt} \right|_{\text{sat}} = - \frac{\sigma(\kappa_s(q_v - q_s)) \cdot (q_v - q_s)}{\Delta t}, \quad (5.24)$$

with sigmoid sharpness κ_s . This ensures all microphysics outputs have consistent units of $\text{kg kg}^{-1} \text{s}^{-1}$.

Autoconversion (cloud $\rightarrow$ rain).

$$\left. \frac{dq_c}{dt} \right|_{\text{auto}} = -k_{\text{auto}} \text{softplus}(q_c - q_{c,\text{thresh}}), \quad k_{\text{auto}} = 10^{-3} \text{ s}^{-1}, \quad q_{c,\text{thresh}} = 10^{-3} \text{ kg/kg}. \quad (5.25)$$

Accretion (collection of cloud by rain).

$$\left. \frac{dq_c}{dt} \right|_{\text{accr}} = -k_{\text{accr}} q_c q_r^{0.875}, \quad k_{\text{accr}} = 2.2. \quad (5.26)$$

Rain evaporation.

$$\left. \frac{dq_r}{dt} \right|_{\text{evap}} = k_{\text{evap}} \frac{q_s - q_v}{q_s} q_r^{0.525}. \quad (5.27)$$

Sedimentation. Fall speed: $V_t = V_0 \sqrt{\rho_{\text{sfc}}/\rho}$ with $V_0 = 5$ m/s.

Latent heating. $dT/dt = (L_v/c_p) \cdot (\text{condensation} - \text{evaporation})$.

5.4.2 Seifert–Beheng Two-Moment Warm Rain

Prognostic mass (q_c, q_r) and number concentration (N_c, N_r).

Autoconversion.

$$\left. \frac{dq_c}{dt} \right|_{\text{auto}} = -k_{\text{au}} q_c^2 \sigma[\kappa(x_c - x^*)], \quad x_c = \frac{q_c \rho}{N_c}, \quad (5.28)$$

where $k_{\text{au}} = 600 \text{ m}^3/(\text{kg s})$, $x^* = 2.6 \times 10^{-10}$ kg is the separation mass, and the sigmoid onset replaces the hard threshold.

Self-collection and breakup.

$$\left. \frac{dN_r}{dt} \right|_{\text{sc}} = -k_{\text{sc}} N_r q_r \rho, \quad \left. \frac{dN_r}{dt} \right|_{\text{br}} = +k_{\text{br}} \sigma[\kappa(D_r - D_{\text{eq}})] \left| \left. \frac{dN_r}{dt} \right|_{\text{sc}} \right|. \quad (5.29)$$

5.4.3 Morrison Double-Moment (Ice + Liquid)

Extends the warm-rain scheme with ice-phase processes:

Ice nucleation (Cooper 1986).

$$N_{i,\text{target}} = \frac{N_{i,0}}{\rho} \exp[a_C(T_{\text{freeze}} - T)]. \quad (5.30)$$

Depositional growth.

$$\left. \frac{dq_i}{dt} \right|_{\text{dep}} = k_{\text{dep}} \text{softplus}(S_i) \sqrt{N_i}, \quad S_i = \frac{q_v}{q_{s,i}} - 1. \quad (5.31)$$

Bergeron process (mixed-phase). Cloud water converts to ice in the temperature window $T_{\text{center}} - T_{\text{width}} < T < T_{\text{freeze}}$:

$$\left. \frac{dq_c}{dt} \right|_{\text{Berg}} = -k_{\text{Berg}} q_c f_{\text{ice}}(T). \quad (5.32)$$

Riming, aggregation, melting. Riming: $q_c \rightarrow q_i$ and $q_c \rightarrow q_s$ via collection. Aggregation: $q_i \rightarrow q_s$ when crystals grow large enough. Melting: sigmoid-activated near T_{freeze} .

Fall speeds (power law).

$$V_t = a \left(\frac{\rho q}{\rho_{\text{sfc}}} \right)^b, \quad \text{rain: } a = 130, b = 0.5; \quad \text{ice: } a = 50, b = 0.25; \quad \text{snow: } a = 30, b = 0.3. \quad (5.33)$$

5.4.4 Thompson Hybrid-Moment Microphysics

Extends Morrison with graupel. Rimed ice exceeding a critical threshold converts to graupel:

$$\left. \frac{dq_g}{dt} \right|_{\text{rime}} = f_g \left. \frac{dq_i}{dt} \right|_{\text{rime}} \cdot \sigma \left[\kappa \left(\left. \frac{dq_i}{dt} \right|_{\text{rime}} - R_{\text{thresh}} \right) \right]. \quad (5.34)$$

5.4.5 Summary of Microphysics Schemes

Scheme	Moments	Species	Key features
Kessler	1	q_c, q_r	Warm rain only
Seifert–Beheng	2	q_c, q_r, N_c, N_r	Realistic droplet spectra
Morrison	2	$q_c, q_r, q_i, q_s, N_c, N_r, N_i$	Full mixed-phase
Thompson	Hybrid	$+q_g$	Graupel, riming

5.5 Cloud Fraction

5.5.1 Sundqvist (1988)

Diagnostic cloud fraction based on a relative humidity threshold:

$$c_f = \text{clamp} \left(\frac{\text{RH} - \text{RH}_{\text{crit}}}{1 - \text{RH}_{\text{crit}}}, 0, 1 \right), \quad \text{RH}_{\text{crit}} = 0.8. \quad (5.35)$$

5.5.2 Xu–Randall (1996)

Combines relative humidity and condensate amount:

$$c_f = \text{RH}^p \left[1 - \exp \left(- \frac{\alpha q_c}{(1 - \text{RH}) q_s} \right) \right], \quad p = 2, \alpha = 100. \quad (5.36)$$

Scheme dispatch. `cloud_fraction.compute_cloud_properties` dispatches on `config.scheme` and accepts only "sundqvist" or "xu_randall"; any other value raises `ValueError`. Disabling cloud diagnostics is handled *upstream* by setting `cloud_scheme="none"` on the caller's config so that `compute_cloud_properties` is not invoked at all — passing "none" to this function directly is itself an error. Earlier versions silently fell back to Sundqvist for any unrecognised string, masking config typos and dead branches; this is now uniform with the rest of the `integration.py` factory contract.

5.6 Boundary-Layer Turbulence

5.6.1 Louis (1979) Local Closure

Mixing Length

$$l = \frac{\kappa_{\text{vK}} z}{1 + \kappa_{\text{vK}} z/l_{\text{max}}}, \quad \kappa_{\text{vK}} = 0.4, \quad l_{\text{max}} = 300 \text{ m}. \quad (5.37)$$

Richardson Number

$$\text{Ri} = \frac{N^2}{S^2}, \quad N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}, \quad S^2 = \left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2. \quad (5.38)$$

Stability Functions

$$f_m(\text{Ri}) = \begin{cases} 1 - \frac{2b \text{ Ri}}{1 + 3bc l^2 |\text{Ri}|^{1/2} / \Delta z^2} & \text{Ri} < 0, \\ \frac{1}{1 + 2b \text{ Ri} / \sqrt{1 + d \text{ Ri}}} & \text{Ri} \geq 0, \end{cases} \quad (5.39)$$

with $b = c = d = 5$. Eddy diffusivities: $K_m = l^2 S f_m$, $K_h = K_m$.

5.6.2 Holtslag–Boville Nonlocal K-Profile

PBL Height Diagnosis

Bulk Richardson number:

$$\text{Ri}_{\text{bulk}}(z) = \frac{g}{\theta_{v,\text{sfc}}} \frac{(\theta_v - \theta_{v,\text{sfc}}) z}{|\mathbf{V}|^2}. \quad (5.40)$$

PBL height h is diagnosed where Ri_{bulk} crosses $\text{Ri}_{\text{crit}} = 0.25$, using sigmoid-weighted averaging for a smooth, differentiable diagnosis.

K-Profile Inside PBL

$$K_m = \kappa_{\text{vK}} u_* z (1 - z/h)^2. \quad (5.41)$$

Above the PBL, Louis (1979) Ri-dependent mixing applies.

5.6.3 TKE-Based Closure

Prognostic turbulent kinetic energy:

$$\frac{\partial \text{TKE}}{\partial t} = P - \varepsilon + D + B, \quad (5.42)$$

where P is shear production, $\varepsilon = c_\varepsilon \text{TKE}^{3/2}/l$ is dissipation, D is diffusion, and B is buoyancy production/destruction. Closure: $K_m = c_m \sqrt{\text{TKE}} \cdot l$, $K_h = K_m / \text{Pr}_t$.

5.6.4 Implicit Vertical Diffusion

The backward-Euler tridiagonal system $-a_k \varphi_{k-1} + b_k \varphi_k - c_k \varphi_{k+1} = d_k$ is solved with the Thomas algorithm:

$$a_k = \frac{\Delta t K_{k-1/2}}{\rho_k \Delta z_k \Delta z_{k-1/2}}, \quad c_k = \frac{\Delta t K_{k+1/2}}{\rho_k \Delta z_k \Delta z_{k+1/2}}, \quad b_k = 1 + a_k + c_k. \quad (5.43)$$

5.7 Surface Exchange Formulations

5.7.1 Neutral Bulk Aerodynamic Fluxes

$$\tau_x = -\rho C_D |\mathbf{V}| u, \quad \tau_y = -\rho C_D |\mathbf{V}| v, \quad (5.44)$$

$$\text{SH} = \rho c_p C_H |\mathbf{V}| (T_s - T_a), \quad \text{LH} = \rho L_v C_E |\mathbf{V}| (q_s - q_a), \quad (5.45)$$

with $C_D = C_H = C_E = 1.5 \times 10^{-3}$.

5.7.2 Monin–Obukhov Similarity Theory (MOST)

Obukhov length.

$$L = -\frac{u_*^3 T_v}{\kappa_{\text{vK}} g (\theta_* + 0.61 T_a q_*)}. \quad (5.46)$$

Businger–Dyer stability functions.

$$\psi_m(\zeta) = \begin{cases} 2 \ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - 2 \arctan x + \frac{\pi}{2}, & \zeta < 0, \quad x = (1 - 16\zeta)^{1/4}, \\ -5\zeta, & \zeta \geq 0, \end{cases} \quad (5.47)$$

$$\psi_h(\zeta) = \begin{cases} 2 \ln\left(\frac{1+y}{2}\right), & \zeta < 0, \quad y = (1 - 16\zeta)^{1/2}, \\ -5\zeta, & \zeta \geq 0. \end{cases} \quad (5.48)$$

Transfer coefficients.

$$C_D = \left[\frac{\kappa_{\text{vK}}}{\ln(z/z_0) - \psi_m(\zeta)} \right]^2, \quad C_H = \frac{\kappa_{\text{vK}}^2}{[\ln(z/z_0) - \psi_m(\zeta)][\ln(z/z_{0t}) - \psi_h(\zeta)]}. \quad (5.49)$$

The iterative solver (5 iterations via `jax.lax.fori_loop`) is fully differentiable.

5.7.3 COARE 3.0 Roughness

Over ocean (Fairall et al., 2003):

$$z_0 = \alpha_c \frac{u_*^2}{g} + 0.11 \frac{\nu}{u_*}, \quad z_{0t} = \min(1.15 \times 10^{-4}, 5.5 \times 10^{-5} \text{Re}_*^{-0.6}), \quad (5.50)$$

where $\alpha_c = 0.011$ is the Charnock parameter.

5.7.4 Large & Yeager (2004) Roughness

$$C_{DN} \times 10^3 = \frac{2.7}{U_{10N}} + 0.142 + 0.0764 U_{10N}. \quad (5.51)$$

5.8 Gravity Wave Drag

5.8.1 Orographic GWD (Lindzen 1981)

Surface stress.

$$\tau_0 = \rho N k h_{\text{topo}}^2 U, \quad N = \sqrt{\frac{g}{\theta} \frac{d\theta}{dz}}. \quad (5.52)$$

Saturation criterion. Waves break when the Froude number $\text{Fr} = U/(N h_{\text{topo}})$ drops below a critical value. Stress is capped at the saturation value $\tau_{\text{sat}} = \rho |U|^3 k/N$.

Momentum deposition.

$$\left. \frac{du}{dt} \right|_{\text{GWD}} = -\frac{1}{\rho} \frac{d\tau}{dz}. \quad (5.53)$$

Frictional heating.

$$\left. \frac{dT}{dt} \right|_{\text{GWD}} = -\frac{1}{c_p} \left(u \frac{du}{dt} + v \frac{dv}{dt} \right). \quad (5.54)$$

5.9 Held–Suarez Forcing

The idealised forcing benchmark of [Held and Suarez \(1994\)](#):

Equilibrium temperature.

$$T_{\text{eq}} = \max\left(200, \left[315 - \Delta T_y \sin^2 \phi - \Delta \theta_z \ln(p/p_0) \cos^2 \phi\right] (p/p_0)^\kappa\right), \quad (5.55)$$

with $\Delta T_y = 60$ K, $\Delta \theta_z = 10$ K.

Newtonian relaxation.

$$\frac{dT}{dt} = -k_T(\sigma, \phi) (T - T_{\text{eq}}), \quad k_T = k_a + (k_s - k_a) \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right) \cos^4 \phi, \quad (5.56)$$

with $k_a = 1/(40 \text{ day})$, $k_s = 1/(4 \text{ day})$, $\sigma_b = 0.7$.

Rayleigh friction.

$$\frac{dv}{dt} = -k_v(\sigma) \mathbf{v}, \quad k_v = k_f \max\left(0, \frac{\sigma - \sigma_b}{1 - \sigma_b}\right), \quad k_f = 1/(1 \text{ day}). \quad (5.57)$$

5.10 Additional Turbulence Schemes

5.10.1 Smagorinsky

Constant eddy diffusivity: $K_m = K_h = \text{const.}$

5.10.2 YSU (Yonsei University)

Nonlocal K-profile boundary-layer scheme with explicit entrainment at the PBL top, following [Holtslag and Boville \(1993\)](#) with enhanced counter-gradient terms.

5.10.3 CLUBB-lite

Higher-order prognostic moment closure with five variables: $\overline{w'^2}$, $\overline{\theta_l'^2}$, $\overline{r_t'^2}$, $\overline{w'\theta_l'}$, $\overline{w'r_t'}$. Cloud fraction is diagnosed via an assumed bivariate Gaussian PDF of θ_l and r_t .

5.10.4 EDMF Turbulence

Eddy-diffusivity mass-flux unified boundary-layer framework combining diffusive transport (for weakly convective conditions) with explicit updraft mass flux (for strongly convective conditions) in a single scheme.

5.11 Additional Convection Schemes

5.11.1 Kuo (1974)

Column moisture-excess scheme activating when column-integrated moisture convergence exceeds a threshold. Excess moisture is partitioned between heating (fraction α_{heat}) and moistening ($1 - \alpha_{\text{heat}}$).

5.11.2 Deep Convective Adjustment (DCA)

Bottom-to-top column adjustment toward a moist-adiabatic reference profile, with smooth sigmoid activation for JAX differentiability.

5.12 Gravity Wave Drag Schemes

Six GWD parameterisations are available:

1. **Rayleigh friction:** Linear damping above a threshold height.
2. **Lindzen (1981):** Orographic GWD with saturation and critical-level absorption.
3. **McFarlane (1987):** Enhanced orographic scheme with improved wave saturation.
4. **Hines (1997):** Doppler-spread parameterisation for non-orographic GWD, propagating a spectrum of gravity waves upward with nonlinear interactions.
5. **Prognostic Spectral:** Multi-azimuthal (N_{az}), multi-wavenumber (N_k) spectrum with prognostic wave momentum flux and vertical propagation via `jax.lax.scan`.
6. **ML Emulator:** Equinox MLP surrogate mapping column state to GWD tendencies.

5.13 Neural and Learned Physics

5.13.1 NeuralPhysics

An Equinox MLP column model that maps the column state (temperature, moisture, wind, surface fluxes) to physics tendencies. Hybrid mode blends learned corrections with conventional parameterisation output via a configurable blending coefficient.

5.13.2 LearnedColumn

Per-column physics replacement following Rasp et al. (2018), mapping the full column state vector to temperature and moisture tendencies via a neural network, trainable end-to-end with the dynamical core.

Chapter 6

Ocean Model

6.1 Boussinesq Hydrostatic Primitive Equations

In vector-invariant form on the cubed sphere:

$$\frac{\partial u}{\partial t} = (\zeta + f) v - \frac{\partial K}{\partial x} - \frac{1}{\rho_0} \frac{\partial p'}{\partial x} + A_h \nabla^2 u + \frac{\partial}{\partial z} \left(A_v \frac{\partial u}{\partial z} \right) - w \frac{\partial u}{\partial z}, \quad (6.1)$$

$$\frac{\partial v}{\partial t} = -(\zeta + f) u - \frac{\partial K}{\partial y} - \frac{1}{\rho_0} \frac{\partial p'}{\partial y} + A_h \nabla^2 v + \frac{\partial}{\partial z} \left(A_v \frac{\partial v}{\partial z} \right) - w \frac{\partial v}{\partial z}. \quad (6.2)$$

The Coriolis force is split: depth-mean $f\mathbf{v}$ in the barotropic substeps, baroclinic shear $f\mathbf{v}'$ in the slow tendencies.

6.1.1 Tracer Equations (Skew-Symmetric)

$$\frac{\partial \phi}{\partial t} = \frac{1}{2} \left[-\mathbf{v} \cdot \nabla \phi - \nabla \cdot (\phi \mathbf{v}) \right] - w \frac{\partial \phi}{\partial z} + K_h \nabla^2 \phi + \frac{\partial}{\partial z} \left(K_v \frac{\partial \phi}{\partial z} \right). \quad (6.3)$$

6.1.2 Free-Surface Equation

$$\frac{\partial \eta}{\partial t} = - \sum_k \nabla \cdot (h_k \mathbf{v}_k). \quad (6.4)$$

6.2 Barotropic Solver

Split-explicit forward-backward substeps:

$$\frac{\partial \eta}{\partial t} = -\nabla \cdot (H \bar{\mathbf{V}}), \quad \frac{\partial \bar{\mathbf{V}}}{\partial t} = f \hat{\mathbf{k}} \times \bar{\mathbf{V}} - g \nabla \eta + \mathbf{F}_{\text{slow}}. \quad (6.5)$$

Semi-implicit Coriolis. With $\alpha = \frac{1}{2} f \Delta t_s$:

$$\bar{U}^{n+1} = \frac{R_u + \alpha R_v}{1 + \alpha^2}, \quad \bar{V}^{n+1} = \frac{R_v - \alpha R_u}{1 + \alpha^2}, \quad (6.6)$$

where $R_u = \bar{U}^n + \alpha \bar{V}^n - \Delta t_s g \partial \eta / \partial x$ and $R_v = \bar{V}^n - \alpha \bar{U}^n - \Delta t_s g \partial \eta / \partial y$.

6.3 Equation of State (Wright 1997)

The ocean prognostic temperature variable T is **potential temperature** [degC]. The Wright (1997) EOS computes in-situ density $\rho(\theta, S, p)$ directly from potential temperature θ , salinity S [PSU], and pressure p [Pa]:

$$\rho = \frac{p + p_0(\theta, S)}{\lambda(\theta, S) + \alpha_0(\theta, S)(p + p_0)}, \quad (6.7)$$

where

$$\alpha_0 = a_0 + a_1\theta + a_2S, \quad (6.8)$$

$$p_0 = (b_0 + b_4S) + \theta(b_1 + \theta(b_2 + b_3\theta) + b_5S), \quad (6.9)$$

$$\lambda = (c_0 + c_4S) + \theta(c_1 + \theta(c_2 + c_3\theta) + c_5S). \quad (6.10)$$

Coefficients follow the MOM6 implementation. The density perturbation $\rho' = \rho(\theta, S, p) - \rho_0$ is used in the pressure gradient computation.

6.4 Hydrostatic Pressure and Vertical Velocity

Baroclinic pressure by downward integration:

$$p'_k = \rho_0 g \eta + \sum_{k'=0}^{k-1} \rho'_{k'} g \Delta z_{k'} + \frac{1}{2} \rho'_k g \Delta z_k. \quad (6.11)$$

Vertical velocity diagnosed bottom-up from $w(\text{floor}) = 0$:

$$w_{k-1/2} = - \sum_{k'=k}^{N-1} \nabla \cdot (h_{k'} \mathbf{v}_{k'}). \quad (6.12)$$

6.5 Vertical Mixing: K-Profile Parameterisation (KPP)

Following [Large et al. \(1994\)](#):

6.5.1 Boundary Layer Depth

The boundary layer depth h is diagnosed where the bulk Richardson number crosses $\text{Ri}_{\text{crit}} = 0.25$ (LMD94 default):

$$\text{Ri}_b(z) = \frac{g \Delta \rho z}{\rho_0 (|\Delta \mathbf{V}|^2 + V_t^2)}, \quad (6.13)$$

using sigmoid-weighted averaging for a differentiable diagnosis.

6.5.2 K-Profile Shape

$$K_m(\sigma) = h w_s(\sigma) G(\sigma), \quad G(\sigma) = \sigma(1 - \sigma)^2, \quad \sigma = z/h, \quad (6.14)$$

where w_s is the turbulent velocity scale depending on surface fluxes and stability.

6.5.3 Non-Local Transport

For unstable (convective) boundary layers ($B_f > 0$), a non-local counter-gradient flux is added to temperature and salinity tendencies. The implementation uses a simplified form inspired by LMD94 Eq. 19:

$$\left. \frac{\partial T}{\partial t} \right|_{\text{NL}} = \gamma_T \frac{w_s(0) G(\sigma)}{w_s(\sigma) h} \left. \frac{\partial T}{\partial z} \right|_{\text{sfc}}, \quad (6.15)$$

where $\gamma_T = \gamma_S = 6.33$ are non-local transport coefficients and $w_s(0) (\partial T / \partial z)|_{\text{sfc}}$ is a surface heat flux proxy [K m/s]. This differs from the full LMD94 non-local term, which uses the explicit surface buoyancy flux $\overline{w'\theta'}_0$; the present form approximates it via the near-surface gradient and surface turbulent velocity scale. This is active only within the diagnosed boundary layer ($\sigma < 1$) and only under unstable forcing.

6.6 Lateral Mixing

6.6.1 Harmonic (Laplacian) Diffusion

$$D_h = A_h \nabla^2 \phi. \quad (6.16)$$

6.6.2 Biharmonic (∇^4) Diffusion

$$D_h = -A_4 \nabla^4 \phi. \quad (6.17)$$

6.6.3 Gent–McWilliams / Redi Parameterisation

The GM parameterisation represents the effect of mesoscale eddies on tracer transport via an additional advective flux (bolus velocity) ([Gent and McWilliams, 1990](#)):

$$\mathbf{u}^* = -\kappa_{\text{GM}} \nabla_h s, \quad (6.18)$$

where $s = -\nabla_h \rho / (\partial \rho / \partial z)$ is the isopycnal slope. The Redi diffusion tensor rotates lateral mixing along isopycnal surfaces.

6.7 Tracer Advection Schemes

The *legoESM* ocean carries a family of flux-form tracer advection schemes of increasing accuracy in `src/legoesm/ocean/advection.py` and `advection_som.py`. Following [Hill et al. \(2012\)](#), implicit numerical mixing is the dominant source of spurious diapycnal transport in z -coordinate models, so the volume-weighted variance $\text{Var}(T) = \int (T - \langle T \rangle)^2 dV$ is used as the definitive evaluation metric (it is an exact inviscid invariant in closed-domain zero- K_h Eady setups).

Scheme inventory.

Scheme	Order / halo	Notes
<code>upwind</code>	1, $h=1$	Donor-cell; most diffusive but unconditionally stable.
<code>tvd</code>	2, $h=2$	Van Leer / Sweby slope-limited flux; total-variation diminishing.
<code>dst3</code>	3, $h=2$	Direct-space-time third-order, used in vertical advection.
<code>ppm</code>	4, $h=2$	Parabolic Piecewise Method (Colella–Woodward 1984).
<code>ppm.fct</code>	4, $h=2$	PPM with Zalesak FCT limiter (sign-split limiter pending, #9).
<code>weno5/weno7</code>	5/7, $h=3/4$	WENO smoothness-weighted reconstructions.
<code>som</code>	Prather 9-moment	Second-Order Moments; advects all 9 sub-cell moments per tracer.

SOM (Prather 1986). The SOM solver in `advection.som.py` carries the cell mean and nine sub-cell moments $\{S_x, S_y, S_z, S_{xx}, S_{yy}, S_{zz}, S_{xy}, S_{xz}, S_{yz}\}$, performs operator-split donor/receiver sweeps (`_som_x_sweep`, `_som_y_sweep`, `_som_z_sweep`), and applies moment limiters (`_limit_moments`) to enforce monotonicity. The scheme is unconditionally monotone and conserves all advected moments to machine precision; it is the lowest-numerical-diffusion option in the inventory.

Eady channel benchmark. On the Eady uniform channel (100×50 , 20 levels, no sponge, $B_h = 2.3 \times 10^{11} \text{ m}^4 \text{ s}^{-1}$, $K_h = A_h = 0$):

Scheme	200 d strong ($U=0.8$)	600 d weak ($U=0.2$)	Cost vs. <code>upwind</code>
<code>upwind</code>	86% $\text{Var}(T)$ lost	12% lost	1.0×
<code>tvd</code>	70% lost	6% lost	1.02×
<code>som</code>	blow-up day 107	—	2.8×
<code>som</code> + $C_{\text{Smag}}=0.2$	46% lost	2.9% lost	2.2×
<code>ppm.fct</code>	blow-up day 30	—	$\sim 1.3 \times$ (est.)

The strong-forcing regime saturates at $p_{95}|\text{Ro}| \approx 1.5\text{--}3$ (submesoscale, where diapycnal mixing is physically mandatory); the weak-forcing regime stays at $p_{95}|\text{Ro}| \approx 0.05\text{--}0.11$ (balanced ocean interior), so even `upwind` preserves 88.5% of $\text{Var}(T)$ over 600 days. The *absolute* amount of numerical mixing is dominated by the forcing regime, not the scheme.

Recommendations. Production AMIP / coupled runs default to `tvd` (safe, $\leq 2\%$ overhead); science runs targeting fronts and water-mass tracking use `som` + `Smagorinsky` ($C_{\text{Smag}}=0.2$) which combines nine-moment conservation with just enough momentum dissipation to suppress grid-scale noise. The `ppm.fct` branch is gated on the sign-split Zalesak/Kuzmin limiter (open issue #9). Convergence methodology and 2D/3D test runners are documented in `docs/ocean_experiments/advection_convergence`.

6.8 Slab and Two-Layer Ocean

For idealised experiments, a simple mixed-layer energy balance:

$$C_{\text{mix}} \frac{dT_s}{dt} = \text{SW}_{\text{net}} + \text{LW}_{\text{net}} - \text{SH} - \text{LH} + Q_{\text{flux}} + K_{\text{mix}}(T_{\text{deep}} - T_s), \quad (6.19)$$

where $C_{\text{mix}} = \rho_{\text{ocn}} c_{\text{ocn}} h_{\text{mix}}$ is the mixed-layer heat capacity.

6.9 Realistic-Geometry Spinup on ETOPO Bathymetry

The lat-lon C-grid ocean supports global Wolfe–Cessi-style spinups on real ETOPO/GEBCO bathymetry through a topology-aware loader and a set of stability-conditioning operators specific to coastal partial cells and high latitudes.

Bathymetry pre-processing. `src/legoesm/ocean/bathymetry.py` exposes `BathymetryConfig` and the per-grid loaders (`load_bathymetry_latlon_cgrid`, `load_bathymetry_cubed_sphere`, `load_bathymetry_mp`, `load_bathymetry_gaussian`). The pipeline is:

1. Auto-detect the depth/elevation variable across ETOPO, GEBCO, NOAA ERDDAP, and generic CF conventions.
2. Regrid to the model grid (conservative or bilinear) and clip to $H \in [H_{\text{min}}, H_{\text{max}}]$ with $H_{\text{min}} = 50$ m by default (preserves mid-ocean ridges).
3. Optional Laplacian smoothing on the cubed-sphere or Voronoi mesh (`_laplacian_smooth_2d`, `_laplacian_smooth_voronoi`).
4. Steepness-cap via the Mellor–Ezer–Oey iteration `apply_meo_r_factor_cap` which enforces $r_{\text{max}} = |\Delta H|/(2\bar{H}) \leq r_{\text{cap}}$ (typical $r_{\text{cap}} = 0.2$); the per-call `initial_r_max` / `final_r_max` are logged for the diagnostic scripts.
5. Optional polar-cap mask (`north_cap_lat`, `south_cap_lat`) and `enforce_straits` to thread Drake / Indonesian throughflow at low resolution.
6. Isolated-basin filling (`_fill_isolated_basins`) so disconnected pockets do not trap tracers.

AL81 partial-cell Coriolis. The vector-invariant momentum branch uses an Arakawa–Lamb 1981 (“EEN”) 12-corner-triad PV-flux closure (`pv_flux_al81_partial_cell` in `latlon_cgrid_operators.py`). Each f -point carries four 1/12-weighted L-shaped triads of vertex potential vorticity $q = \zeta/h_{\text{vtx}}$; the partial-cell extension after [Le Sommer et al. \(2009\)](#) weights each triad by the local wet-cell thickness so that discrete potential enstrophy is preserved across bottom steps. Coastal triads use a Neumann fill of q (three-pass smoothing band-aid pending the formal mask-projection closure documented in `docs/ocean_experiments/al81_corner_triad_audit.md`).

Biharmonic viscosity scaling. Because lat-lon zonal spacing scales as $\cos \phi$, a constant biharmonic coefficient B_h violates CFL near the poles while under-diffusing at the equator. `biharmonic_scaling_factor` multiplies B_h by $\cos^4 \phi$ at u - and v -faces following the MOM6 convention of [Griffies and Hallberg \(2000\)](#), so the effective diffusivity tracks dx_{local}^4 uniformly. This is the dominant lever for suppressing the alternating-sign $2\Delta y$ zonal-jet computational mode that develops at the equator on partial cells.

Validated configuration. The 20-yr `global_overturning_realistic_geometry` run (branch `realistic-geometry-full`, PR #224) uses $A_h = 10^4 \text{ m}^2 \text{ s}^{-1}$, $B_h = 5 \times 10^9 \text{ m}^4 \text{ s}^{-1}$, $\kappa_{\text{GM}} = \kappa_{\text{Redi}} = 800 \text{ m}^2 \text{ s}^{-1}$, SMC03 pressure-gradient force, AL81 vorticity advection, implicit Crank–Nicolson barotropic solver, and bottom-drag $r = 2.5 \times 10^{-3}$. Stratification reproduces the canonical thermohaline structure (warm pool aloft, cold abyss, sloping isotherms, warm-equator/cold-pole SST imprint). Surface speeds remain inflated $\sim 3\text{--}5\times$ above realistic values from residual partial-cell q -stencil noise; the follow-up plan is the full Sadourny–Salmon energy–enstrophy form with all bottom-cell ENE corrections. See `docs/ocean_experiments/realistic_geometry_phase4_results.md` for the year-by-year diagnostics.

Chapter 7

Ocean Biogeochemistry

7.1 Carbonate Chemistry and Air–Sea CO₂ Exchange

7.1.1 Gas Transfer Velocity

Following [Wanninkhof \(2014\)](#):

$$k_w = 0.251 U_{10}^2 \left(\frac{\text{Sc}}{660} \right)^{-1/2}, \quad (7.1)$$

where the Schmidt number is a polynomial function of temperature:

$$\text{Sc} = 2116.8 - 136.25 T + 4.7353 T^2 - 0.092307 T^3 + 7.555 \times 10^{-4} T^4. \quad (7.2)$$

7.1.2 Net Air–Sea Flux

$$F_{\text{CO}_2} = k_w K_0 (p\text{CO}_{2,\text{ocean}} - p\text{CO}_{2,\text{atm}}), \quad (7.3)$$

where K_0 is Henry’s law solubility. The ocean surface $p\text{CO}_2$ is computed from DIC, ALK, T , S via carbonate equilibria using Newton–Raphson iteration.

7.2 NPZD Ecosystem Model

Four biological tracers: NO_3 (nutrients), P (phytoplankton), Z (zooplankton), D (detritus), following [Fasham et al. \(1990\)](#):

$$\frac{\partial P}{\partial t} = \mu_P P - G - m_P P, \quad (7.4)$$

$$\frac{\partial Z}{\partial t} = \gamma_Z G - m_Z Z^2, \quad (7.5)$$

$$\frac{\partial D}{\partial t} = (1 - \gamma_Z)G + m_P P + m_Z Z^2 - r_D D + w_s \frac{\partial D}{\partial z}, \quad (7.6)$$

$$\frac{\partial \text{NO}_3}{\partial t} = -\mu_P P + r_D D. \quad (7.7)$$

Phytoplankton growth.

$$\mu_P = \mu_{\max} f(T) \min(L_N, L_I), \quad L_N = \frac{\text{NO}_3}{k_N + \text{NO}_3}, \quad L_I = 1 - e^{-k_{\text{PAR}} \cdot \text{PAR}}. \quad (7.8)$$

Grazing (Holling III).

$$G = g_{\max} \frac{P^2}{K_g + P^2} Z. \quad (7.9)$$

Redfield stoichiometry ($R_{C:N} = 6.625$) couples biology to DIC/ALK.

7.3 Ocean Bottom Drag

Two formulations:

Linear drag. $\tau_b = -r \mathbf{u}_b$, with configurable coefficient r .

Quadratic drag. $\tau_b = -C_d |\mathbf{u}_b| \mathbf{u}_b$, with $C_d = 2.5 \times 10^{-3}$ (default).

7.4 Ocean Surface Forcing

Three modes are available:

- **Prescribed:** Fixed wind stress (τ_x, τ_y) and heat flux with latitudinal profiles.
- **Restoring:** SST/SSS restoring toward climatology (Haney-style) with cosine latitude temperature profile (25°C at equator, 0°C at poles).
- **Bulk formulas:** COARE-like air–sea fluxes with drag (C_D), Stanton (C_H), and Dalton (C_E) coefficients, reference height, and stability corrections.

7.5 Shortwave Penetration

Subsurface shortwave absorption follows the Jerlov (1968) water type classification with five types (I, IA, IB, II, III). Each type defines a two-band exponential decay profile:

$$I(z) = I_0 [R e^{-z/\zeta_1} + (1 - R) e^{-z/\zeta_2}], \quad (7.10)$$

where R, ζ_1, ζ_2 depend on the water type.

7.6 Ocean Convection

7.6.1 Enhanced Diffusion

Convective adjustment via enhanced diffusivity where $N^2 < 0$: $K_{\text{conv}} = K_{\text{max}}$ in statically unstable regions, with smooth transitions.

7.6.2 Plume Model

Entraining mass-flux ocean convection with entrainment rate $\alpha_{\text{plume}} = 0.1$ and background detrainment $\varepsilon = 10^{-3} \text{ m}^{-1}$.

7.7 Richardson-Number Vertical Mixing

Interior shear-driven diffusivity:

$$K_v = K_{v,0} (1 + 5 \text{ Ri})^{-1}, \quad \text{Ri} = \frac{N^2}{S^2}, \quad (7.11)$$

complementing KPP boundary-layer mixing with interior shear instability parameterisation.

Chapter 8

Land Surface Model

8.1 Slab Land

8.1.1 Surface Energy Balance

$$C_s d_s \frac{dT_s}{dt} = (1 - \alpha_l) SW^\downarrow + \varepsilon_l (LW^\downarrow - \sigma_{SB} T_s^4) - SH - LH. \quad (8.1)$$

8.1.2 Bucket Hydrology

$$\frac{dW}{dt} = P - E, \quad W \in [0, W_{\max}]. \quad (8.2)$$

Moisture availability $\beta = \beta_{\min} + (1 - \beta_{\min}) W/W_{\max}$ modulates evaporation via $q_{\text{sfc}} = \beta q_s(T_s)$.

8.2 Multi-Layer Land Model

8.2.1 Soil Thermal Diffusion

$$C_{\text{eff}}(z) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[k_{\text{eff}}(z) \frac{\partial T}{\partial z} \right], \quad (8.3)$$

with effective heat capacity:

$$C_{\text{eff}} = (1 - \theta_{\text{sat}}) C_{\text{soil}} + \theta C_{\text{water}} + (\theta_{\text{sat}} - \theta) C_{\text{air}}, \quad (8.4)$$

and thermal conductivity following the [Johansen \(1975\)](#) parameterisation via the Kersten number. Solved implicitly using the Thomas algorithm with surface heat flux and geothermal flux boundary conditions.

8.2.2 Richards Equation (Soil Hydraulics)

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K(\psi) \left(\frac{\partial \psi}{\partial z} + 1 \right) \right], \quad (8.5)$$

where θ is volumetric water content, ψ is matric potential, and $K(\psi)$ is hydraulic conductivity.

Six water retention curves are available:

Curve	$\psi(\theta)$
Van Genuchten (1980)	$\psi = \alpha^{-1}[(S_e^{-1/m} - 1)^{1/n}]$
Campbell (1974)	$\psi = \psi_s(S_e)^{-b}$
Brooks–Corey (1964)	$\psi = \psi_d(S_e)^{-1/\lambda}$
Clapp–Hornberger (1978)	Power-law form

where $S_e = (\theta - \theta_r)/(\theta_{\text{sat}} - \theta_r)$ is effective saturation.

Root water uptake. Root distribution: $f_{\text{root}}(z) = \exp(-z/z_{\text{root}})$, with default $z_{\text{root}} = 0.5$ m. Plant-available water between wilting point (θ_{wp}) and field capacity (θ_{fc}).

8.2.3 Snow Budget

$$\frac{dh_{\text{swe}}}{dt} = P_{\text{snow}} - M, \quad M = \max\left(0, \frac{Q_{\text{net}}}{L_f \rho_w}\right), \quad (8.6)$$

with age-dependent albedo: $\alpha_{\text{snow}}(t) = \alpha_{\text{max}} e^{-t/\tau_{\text{age}}} + \alpha_{\text{min}}(1 - e^{-t/\tau_{\text{age}}})$.

8.3 Land Carbon Cycle

8.3.1 DALEC-990 Six-Pool Model

Six prognostic pools: labile C_{lab} , foliage C_f , root C_r , wood C_w , litter C_{lit} , and soil organic matter C_{SOM} .

Gross Primary Production (GPP).

$$\text{GPP} = \varepsilon \text{APAR} f_T f_{\text{CO}_2} \beta, \quad \text{APAR} = f_{\text{APAR}} \text{PAR}, \quad f_{\text{APAR}} = 1 - e^{-k_{\text{ext}} \text{LAI}}, \quad (8.7)$$

where ε is light-use efficiency, $f_T = \exp[-0.5((T - T_{\text{opt}})/T_{\text{width}})^2]$ is the Gaussian temperature response, and $f_{\text{CO}_2} = \text{CO}_2/(K_{\text{CO}_2} + \text{CO}_2)$ is Michaelis–Menten CO_2 response.

Soil respiration. Exponential Q_{10} temperature sensitivity with moisture limitation.

8.3.2 Farquhar (1980) Photosynthesis

Assimilation rate:

$$A = \min(W_c, W_j) - R_d, \quad (8.8)$$

where:

$$W_c = V_{c,\text{max}} \frac{C_i - \Gamma^*}{C_i + K_c(1 + O/K_o)}, \quad (\text{Rubisco-limited}), \quad (8.9)$$

$$W_j = \frac{J(C_i - \Gamma^*)}{4(C_i + 2\Gamma^*)}, \quad (\text{light-limited}). \quad (8.10)$$

$V_{c,\text{max}}$ and J_{max} follow peaked Arrhenius temperature responses (Bernacchi et al., 2001). The minimum is implemented as a smooth (differentiable) function.

8.3.3 Stomatal Conductance

Three models couple stomata to photosynthesis:

Ball–Berry (1987). $g_s = g_0 + g_1 A \text{RH}/C_s$.

Medlyn (2011). $g_s = g_0 + 1.6(1 + g_1/\sqrt{\text{VPD}}) A/C_s$.

Jarvis (1976). $g_s = g_{s,\max} f(\text{PAR}) f(T) f(\text{VPD}) f(\theta)$ (CO_2 -independent; for when the carbon cycle is inactive).

The coupled A – g_s – C_i system is solved by 5 fixed-point iterations, unrolled for JAX JIT compatibility.

8.4 Plant Functional Types

Land surface parameters are spatially heterogeneous and depend on vegetation type. legoESM implements a CLM5-style Plant Functional Type (PFT) system with 17 types spanning needle-leaf/broadleaf trees (evergreen and deciduous, temperate and tropical), shrubs, grasses (C3/C4), and crops.

8.4.1 Parameter Lookup

For each grid cell, the effective parameter $\bar{p}$ is a PFT-fraction-weighted average:

$$\bar{p} = \sum_{k=1}^{17} f_k p_k, \quad (8.11)$$

where f_k is the fractional coverage of PFT k and p_k is the parameter value for that type.

8.4.2 Trainable Parameters

Twelve land surface parameters are trainable per PFT: vegetation albedo (α_{veg}), emissivity (ε), roughness length (z_0), maximum soil water ($W_{\max}$), soil heat capacity (C_{soil}), soil depth (d_{soil}), root depth (z_{root}), wilting point (θ_{wp}), field capacity (θ_{fc}), maximum carboxylation rate ($V_{c,\max,25}$), leaf carbon mass per area (LCMA), and stomatal slope (g_1).

Each parameter is bounded via sigmoid transforms:

$$p = p_{\min} + (p_{\max} - p_{\min}) \sigma(x_{\text{raw}}), \quad (8.12)$$

where x_{raw} is the unconstrained trainable value.

8.4.3 Neural Parameter Provider

An alternative to PFT lookup, a 3-hidden-layer fully connected network (GELU activation, 48 hidden units) maps static land features—cyclic latitude/longitude, one-hot soil type, normalised elevation, precipitation, temperature, and forest age—to the 12 land parameters. Sigmoid output bounds match the PFT physical ranges.

8.5 DifferLand: Differentiable Land Model Training

The DifferLand framework enables end-to-end gradient-based optimisation of land surface and carbon cycle parameters. Three carbon cycle modes are available:

- **none**: Carbon cycle disabled; LAI prescribed.
- **differland**: Prognostic DALEC-990 with trainable allocation fractions, decomposition rates, and phenology parameters.
- **seasonal**: Prescribed seasonal LAI cycle; carbon pools diagnostic only.

8.5.1 Phenology

Leaf emergence and fall follow hemisphere-aware Gaussian windows:

$$f_{\text{on}}(d) = \exp\left[-\frac{(d - d_{\text{Bday}})^2}{2\tau_{\text{on}}^2}\right], \quad f_{\text{off}}(d) = \exp\left[-\frac{(d - d_{\text{Fday}})^2}{2\tau_{\text{off}}^2}\right], \quad (8.13)$$

where d is the day of year, d_{Bday} and d_{Fday} are budburst and leaf-fall dates, and τ controls the transition width.

8.5.2 Decomposition

Litter and SOM decomposition rates follow Q_{10} temperature sensitivity with moisture limitation:

$$k(T, \theta) = k_0 Q_{10}^{(T - T_{\text{ref}})/10} f_w(\theta), \quad (8.14)$$

where f_w is a piecewise-linear function of soil moisture.

8.5.3 NPP Allocation

Net primary production ($\text{NPP} = \text{GPP} - R_a$) is allocated sequentially: foliar $\rightarrow$ labile $\rightarrow$ root $\rightarrow$ wood, with allocation fractions as trainable parameters.

Chapter 9

Sea Ice

9.1 Thermodynamic Slab

9.1.1 Surface Temperature

$$\rho_i c_i \frac{h_{\text{eff}}}{2} \frac{dT_i}{dt} = Q_{\text{sfc}} - \frac{k_i(T_f - T_i)}{h_{\text{eff}}}, \quad h_{\text{eff}} = \max(h_i, h_{\text{min}}). \quad (9.1)$$

9.1.2 Growth and Melt

$$\frac{dh_i}{dt} = \frac{F_{\text{cond}} - F_{\text{ocean}}}{\rho_i L_f}. \quad (9.2)$$

9.1.3 Concentration

$$\frac{dc}{dt} = \begin{cases} (dh/dt)(1 - c)/h_{\text{new}} & \text{growing,} \\ (dh/dt) c/h_{\text{eff}} & \text{melting.} \end{cases} \quad (9.3)$$

9.2 EVP Rheology

Following [Hunke and Dukowicz \(1997\)](#), the ice momentum equation is

$$m \frac{\partial \mathbf{u}_i}{\partial t} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_w - mf \hat{\mathbf{k}} \times \mathbf{u}_i - mg \nabla \eta + \nabla \cdot \boldsymbol{\sigma}, \quad (9.4)$$

where $\boldsymbol{\sigma}$ is the internal stress tensor from an elliptic yield curve with eccentricity $e = 2$. Ice strength:

$$P = P^* h \exp(-C(1 - c)), \quad P^* = 27.5 \text{ kPa}, \quad C = 20. \quad (9.5)$$

The elastic term provides subcycling stability ($N_{\text{evp}} = 120$ substeps).

9.3 Multi-Category Ice Thickness Distribution

For $N_{\text{cat}} > 1$, per-category state (h_k, T_k, c_k) evolves independently under thermodynamics, then [Lipscomb \(2001\)](#) linear remapping redistributes ice across thickness categories, maintaining a well-resolved thickness distribution.

9.4 Free-Drift Dynamics

When `dynamics="free_drift"`, ice velocity is diagnosed from ocean and atmospheric drag:

$$\mathbf{u}_i = C_{\text{ocean}} \mathbf{u}_{\text{ocean}} + \frac{C_{\text{atm}} \rho_{\text{air}}}{\rho_{\text{ice}}} \mathbf{u}_{\text{wind}}, \quad (9.6)$$

with $C_{\text{ocean}} = 5.5 \times 10^{-3}$, $C_{\text{atm}} = 1.3 \times 10^{-3}$.

9.5 Ice Tracer Transport

When `transport="advection"`, ice thickness, concentration, and temperature are advected by the ice velocity field. For multi-category ice, advection is applied independently to each category via `jax.vmap`.

Chapter 10

Tile-Based Coupler

10.1 Surface Flux Blending

Fluxes are area-weighted blends over ocean, land, sea-ice, and lake tiles:

$$X_{\text{blend}} = f_{\text{ocn}} X_{\text{ocn}} + f_{\text{land}} X_{\text{land}} + f_{\text{ice}} X_{\text{ice}} + f_{\text{lake}} X_{\text{lake}}. \quad (10.1)$$

Tile fractions satisfy $\sum f_i = 1$ with consistency checks.

10.2 Ocean Albedo

The Briegleb (1992) zenith-angle-dependent formula:

$$\alpha_{\text{ocn}} = \frac{0.026}{1.1 \mu_0^{1.7} + 0.065} + 0.15 (\mu_0 - 0.1)(\mu_0 - 0.5)(\mu_0 - 1.0), \quad (10.2)$$

where $\mu_0 = \cos \theta_z$.

10.3 Lake Model

A two-layer model with epilimnion (warm, mixed, default depth $h_{\text{epi}} = 5$ m) and hypolimnion (cold, deep, $h_{\text{hypo}} = 20$ m).

10.3.1 Energy Balance

The epilimnion temperature evolves under the surface energy budget:

$$\rho_w c_w h_{\text{epi}} \frac{dT_{\text{epi}}}{dt} = \text{SW}_{\text{net}} + \text{LW}_{\text{net}} - \text{SH} - \text{LH} - F_{\text{mix}}, \quad (10.3)$$

where the mixing flux between the two layers is

$$F_{\text{mix}} = \rho_w c_w k_{\text{eff}} \frac{T_{\text{epi}} - T_{\text{hypo}}}{d_{\text{mid}}}, \quad (10.4)$$

with wind-enhanced effective diffusivity $k_{\text{eff}} = k_{\text{mix}} (1 + \alpha_{\text{wind}} |\mathbf{V}|)$.

10.3.2 Freezing

Lake temperature is clamped at $T_f = 273.15$ K. Bulk flux options: constant coefficients or iterative Monin–Obukhov similarity theory.

Chapter 11

Machine Learning Components

11.1 Spherical Fourier Neural Operator (SFNO)

11.1.1 Architecture

1. **Encoder:** Pointwise linear $n_{\text{in}} \rightarrow n_{\text{emb}}$.
2. N **processor blocks**, each containing:
 - Layer normalisation,
 - SH analysis $\rightarrow$ learned spectral convolution $\rightarrow$ SH synthesis,
 - GELU activation,
 - Pointwise MLP (two layers with expansion factor),
 - Skip connection.
3. **Decoder:** Pointwise linear $n_{\text{emb}} \rightarrow n_{\text{out}}$.
4. **Big-skip residual:** output = input + decoder($\dots$).

11.1.2 Spectral Convolution

In the SH coefficient space:

$$\hat{f}_{nm}^{\text{out}} = W_{nm} \hat{f}_{nm}^{\text{in}}, \quad (11.1)$$

where $W_{nm} \in \mathbb{R}^{n_{\text{emb}} \times n_{\text{emb}}}$ is a learnable weight matrix per wavenumber pair.

The SFNO can serve as a learned dynamical core or physics emulator within the legoESM framework, sharing the same spectral transform infrastructure as the conventional spectral cores.

Chapter 12

Data Assimilation

12.1 Incremental 4D-Var

legoESM provides a fully implemented incremental 4D-Var data assimilation system, leveraging JAX’s automatic differentiation to compute exact adjoints without hand-coded tangent-linear or adjoint models.

12.1.1 Control Vector

The control vector $\mathbf{x}$ comprises the prognostic atmospheric fields:

- Horizontal winds (u, v) ;
- Temperature T ;
- Specific humidity q ;
- Surface pressure p_s .

The control vector specification supports both grid-point and spectral representations, with configurable variable subsets for targeted assimilation.

12.1.2 Cost Function

The cost function minimises the sum of background and observation terms:

$$J(\mathbf{x}) = \frac{1}{2}(\mathbf{x} - \mathbf{x}_b)^T \mathbf{B}^{-1}(\mathbf{x} - \mathbf{x}_b) + \frac{1}{2} \sum_i (\mathcal{H}_i[\mathcal{M}_{t_0 \rightarrow t_i}(\mathbf{x})] - \mathbf{y}_i)^T \mathbf{R}_i^{-1}(\mathcal{H}_i[\mathcal{M}_{t_0 \rightarrow t_i}(\mathbf{x})] - \mathbf{y}_i), \quad (12.1)$$

where $\mathbf{x}_b$ is the background state, $\mathbf{B}$ is the background error covariance, $\mathcal{M}$ is the forward model, $\mathcal{H}_i$ are observation operators, $\mathbf{y}_i$ are observations, and $\mathbf{R}_i$ are observation error covariances.

12.1.3 Background Error Covariance

Four $\mathbf{B}$ matrix families:

- **Diagonal:** $\mathbf{B} = \text{diag}(\sigma_b^2)$;
- **Diffusion:** Implicit diffusion-based correlation modelling;
- **Spectral:** Diagonal in spectral space (scale-dependent variances);
- **Hybrid:** Linear combination of flow-dependent ensemble covariance and static $\mathbf{B}$.

12.1.4 Minimisation

L-BFGS and conjugate gradient solvers with optional $\mathbf{B}^{1/2}$ preconditioning for improved convergence. The gradient $\nabla_{\mathbf{x}}J$ is computed exactly via `jax.grad` through the full forward model trajectory.

12.1.5 Observation Operators

Three observation operator types:

- **Direct:** Identity mapping for in-situ observations on model grid points;
- **Interpolating:** Bilinear/trilinear interpolation from model grid to observation locations;
- **Composite:** Chained operators for derived quantities (e.g., satellite radiances via a forward radiative transfer step).

All operators are JAX-differentiable, so `jax.grad` propagates through observation-space transformations seamlessly.

12.1.6 Analysis–Forecast Cycling

Sequential cycling: the analysis from one assimilation window becomes the background for the next, with checkpoint/restart support.

12.1.7 Incremental Analysis Update

To reduce initialisation shocks, the analysis increment $\delta\mathbf{x} = \mathbf{x}_a - \mathbf{x}_b$ is applied gradually over an IAU window of N_{IAU} timesteps:

$$\mathbf{x}^{n+1} = \mathcal{M}(\mathbf{x}^n) + \frac{\delta\mathbf{x}}{N_{\text{IAU}}}, \quad n = 0, \dots, N_{\text{IAU}} - 1. \quad (12.2)$$

This produces dynamically balanced analyses and is the default cycling mode.

Chapter 13

Conservation and Numerical Corrections

13.1 Atmospheric Mass Conservation

Dry-air mass is corrected by a uniform adjustment to p_s :

$$\Delta\bar{p} = \frac{\sum_i A_i p_{s,i}^{\text{new}} - M_{\text{target}}}{\sum_i A_i}, \quad p_{s,i}^{\text{corr}} = p_{s,i}^{\text{new}} - \Delta\bar{p}. \quad (13.1)$$

Initial-mass anchoring. For long integrations, $M_{\text{target}} = M_0 = \sum_i A_i p_{s,i}^{\text{init}}$ prevents cumulative $O(\varepsilon)$ drift. All accumulations use float64.

Moisture conservation. Column moisture is rescaled: $q^{\text{corr}} = q^{\text{new}} \cdot Q_{\text{old}}/Q_{\text{new}}$.

13.2 Ocean Conservation

Volume: $\eta^{\text{corr}} = \eta^{\text{new}} + (\sum \eta_{\text{old}} A - \sum \eta_{\text{new}} A) / \sum A$.

Heat and salt: uniform additive correction per unit volume. All three grids (cubed-sphere, lat-lon, MPAS) compute corrections simultaneously from the original pre-fix state and upcast to accumulation precision before global reductions.

Note: These are uniform additive fixers that violate local conservation. The proper solution is flux-form tracer advection with barotropic–baroclinic flux reconciliation.

Chapter 14

Differentiable Training

The `training/` module provides end-to-end differentiable training pipelines for three modes:

14.1 Physics Parameter Tuning

`TrainablePhysicsParams` wraps 8 physics parameters (convective relaxation time, mixing length, drag coefficients, etc.) as an Equinox module with sigmoid-based constraints mapping from unconstrained space to valid physical ranges. Gradients flow through `eqx.filter_value_and_grad` applied to the model's `build_segment_fn(...).raw` (non-JIT, non-donating variant for AD compatibility).

14.2 Neural GCM Training

Spectral neural GCM training couples a learned column model with the spectral dynamical core. The loss function combines area-weighted MSE, spectral loss (penalising errors at each total wavenumber), and per-variable MSE, imported from `ml/loss.py`. Optimisation uses `ml/training.create_optimizer` (warmup + cosine decay + gradient clipping).

14.3 SFNO–Dycore Coupling

The Spherical Fourier Neural Operator can be coupled to any dynamical core for hybrid learned–physical simulations. Training uses the same segment-based architecture, with `SegmentForcing` as an explicit argument (not closure-captured) to prevent JIT recompilation when forcing data changes.

14.4 ERA5 Data Ingestion

`era5.to_state.py` converts ERA5 reanalysis (lat-lon) to any model grid with a local Zarr cache for efficient repeated access. Vertical interpolation handles pressure-to-sigma/hybrid coordinate mapping.

Chapter 15

Parallel Runtime

15.1 Execution Modes

Mode	Ranks	Devices/Rank	Halo Backend	Reduction Backend
serial	1	1	local	local
multi_device	1	N	JAX SPMD	local
mpi	N	1	MPI	MPI
hybrid	N	M	MPI + JAX	MPI + JAX

The `ParallelRuntime.create()` entry point detects MPI rank/size, validates device counts, and routes to the correct halo/reduction backend.

15.2 Domain Decomposition

Cubed sphere. Valid device counts: 1, 2, 3, 6, and $6k^2$ for $k \geq 2$. Each face can be further tiled into sub-face patches. `initialize_distributed(global_n=N) → scatter_to_local()` distributes the global state to rank-local patches; `gather_to_global()` is used only for I/O.

Voronoi/MPAS. RCB geometric partitioner (no external dependencies) or METIS graph partitioner. Two-layer halo for biharmonic operators.

15.3 MPI Differentiability

All halo exchange `sendrecv` calls go through a `@jax.custom_vjp` wrapper (`_sendrecv_vjp`) that swaps source and destination ranks in the backward pass, enabling `jax.grad` through distributed halo exchange. Global reductions use AD-safe `allreduce(SUM)` with a custom VJP; `MAX`, `MIN`, `allgather`, and `bcast` are not differentiable and are used only in diagnostics. Conservation fixers (`fix_mass`, `zero_mean_tendency`) correctly flow gradients through global reductions.

The native 4D halo exchange path (`pad_halo_4d`, `pad_halo_vector_4d`) exchanges all vertical levels in one MPI message per neighbour, replacing the earlier per-level `vmap(pad_halo)` which issued n_{lev} separate messages.

15.4 Ensemble Parallelism

`jax.vmap` across ensemble members with `jax.lax.scan` over time steps. Support for perturbed initial conditions, ensemble mean/std/percentile diagnostics, and gradient checkpointing.

15.5 Multi-Device Collective Reductions

The 2026-05 GPU-scaling branch (#228) compresses per-RK3 step collectives across all dycore backends. On the cubed-sphere finite-volume path the four sequential allreduces (one per prognostic variable) collapse to a single fused reduction with shape $(B, h, h, n_{\text{var}})$, cutting halo bandwidth by $\sim 36\text{--}144\times$ at high resolution. The spectral primitive-equation core fuses the FFT and Legendre transforms into a single pass and stacks per-tracer spherical-harmonic analyses; on MPAS, MPI `sendrecv`s drop $12 \rightarrow 6$ per substep and mass-fix payloads $3 \rightarrow 2$. Fifteen SPMD-correctness unit tests gate the changes and ensure SPMD-trace stability under JAX 0.10. Real-hardware GPU/MPI validation on the Levante cluster is described in `docs/REAL_HARDWARE_SCALING.md` and `docs/GPU_SCALING_BRANCH.md`.

Chapter 16

CMIP Infrastructure

16.1 CF/CMOR-Compliant Output

CF-1.8 NetCDF4 files with CMIP6 Data Reference Syntax (DRS) filenames: `<var>.<table>.<model>.<expt>.<var>`
Four CMOR tables: Amon (33 atmospheric variables including clear-sky radiative fluxes, cloud condensate, and relative humidity), Lmon (6 land variables), Omon (2 ocean variables: SST and sea-ice fraction), and Aday (5 daily atmospheric variables) on a standard 19-level pressure grid.

16.2 Experiment Templates

Template	Period	Forcing	CO ₂ range
piControl	1850–∞	Fixed 1850	284.3 ppmv
historical	1850–2014	Transient	284→397 ppmv
ssp245	2015–2100	SSP2-4.5	397→600 ppmv
ssp585	2015–2100	SSP5-8.5	397→1135 ppmv
amip	1979–2014	Prescribed SST	Transient
1pctCO2	1–150	1%/yr increase	284→1277 ppmv

16.3 AMIP CMIP6 Forcing Deck

The CMIP6 AMIP protocol prescribes a six-channel external-forcing stack on top of monthly SST/SIC. legoESM consumes the canonical input4MIPs/MPI-M files directly through six `NamedTuple` configs in `src/legoesm/forcing/external.py`, all of which carry an explicit `start_year` so multi-year files are sampled on their actual calendar timeline rather than collapsed onto a 12-month cyclic axis.

Forcing channels.

Channel	Config (<code>external.py</code>)	Canonical CMIP6/MPI-M file
SST/SIC	AMIPForcingConfig	HadISST/COBE monthly (t, ϕ, λ)
GHG	GHGConfig	greenhouse_historical_plus.nc (annual)
Ozone	OzoneConfig	vmro3.input4MIPs_ozone_CMIP_UReading-CCMI-1-0_*
Solar	SolarConfig	swflux_14band_cmip6_1850-2299-v3.2.nc
Aerosol	AerosolConfig	Kinne aeropt_kinne_sw_b14_*_rast.nc
Volcanic	AerosolConfig.volcanic_*	bc_aeropt_cmip6_volc_lw_b16_sw_b14_*

Time-axis dispatch. For every channel, the loader inspects the file's CF `time:units` and `calendar` attributes via `_load_time_anchor` and maps the simulation day onto the file's absolute axis through `_simday_to_file_day`. Cyclic 12-month climatologies and multi-year transient series are dispatched automatically: ozone files with $n_t > 12$ records preserve interannual evolution; the volcanic file's 12-month layout per year is sampled at the active calendar month so that 1982 (El Chichón) and 1991 (Pinatubo) appear at the correct simulation dates.

GHG annual file. Each of $\{\text{CO}_2, \text{CH}_4, \text{N}_2\text{O}, \text{CFC-11}, \text{CFC-12}\}$ is shape $(n_t, 1, 1)$ with time units "year as %Y.%f". Linear interpolation in fractional years yields scalar concentrations passed to RRTMG as volume mixing ratios via `ghg_concentrations_to_vmr`. Unit scales (1e-6, 1e-9, 1e-12) match input4MIPs convention. GHG concentrations are consumed by RRTMG only — the gray-radiation pathway uses a constant CO_2 optical depth and does not respond to the prescribed time series.

Ozone. O_3 volume mixing ratio $X_{\text{O}_3}(t, \phi, p)$ is loaded with auto-detection of variable name (`vmro3`, `tro3`, `O3`, `ozone`) and unit (mol mol^{-1} vs. kg kg^{-1} auto-converted). The CMIP6 layout (t, p, ϕ, λ) is auto-swapped to (t, ϕ, p) after a zonal mean. Vertical interpolation in `_interp_vertical` handles both ascending and descending pressure axes (CMIP6 files ship $1000 \rightarrow 0.1$ hPa).

Solar TSI and 14-band SW. The MPI-M file provides daily $\text{TSI}(t)$ in W m^{-2} and `SSI_frac` $(t, n_{\text{band}}=14)$ summing to one. When `source="spectral_file"`, the 14 RRTMG-SW bands are expanded to 112 g -points by uniform fill within each band via `_expand_bands_to_gpoints`; the per- g -point fractions are passed into the SW solver to modulate the band-integrated insolation. Solar geometry (zenith angle, declination) is computed analytically and is not part of the prescribed forcing.

Aerosol and volcanic AOD. Background AOD is a zonal-mean monthly climatology $(\text{aod}_{550})(t, \phi)$. Multi-band Kinne files $(t, n_{\text{band}}, \phi, \lambda)$ are reduced to (t, ϕ) by zonal mean and band sum. The volcanic channel adds a second AOD field on top, with optional scaling `volcanic_scale`; both are clipped to $\text{AOD} \geq 0$ before reaching the SW radiation pathway.

Deck driver. `scripts/run_amip_cmip6_deck.py` pins a CMIP6 AMIP physics stack of RRTMG radiation, Sundqvist cloud fraction, Sundqvist large-scale condensation (the deck microphysics default — Kessler in the integrated AMIP path produces NaN winds at day ~ 2 and is tracked as a known issue), SBM convection, and Louis turbulence, and fixes `--ghg-forcing external`, `--ozone-forcing external`, `--solar-source spectral_file`, `--solar-tsi-var TSI`, `--solar-spectral-var SSI_frac`. When forcing files are absent the user must opt in with `--auto-generate` to invoke `scripts/generate_amip_forcing.py` for synthetic CMIP6-shape replacements; otherwise the deck

refuses to run. `scripts/validate_amip_run.py` reads the resulting CMOR output for post-run sanity.

Multi-grid dispatch coverage. `scripts/smoke_test_amip_all_grids.py` exercises the deck *dispatch path* on cubed-sphere (centred / FV3 / C-D), lat-lon (centred / FV3 / C-grid), Gaussian spectral, and Voronoi/MPAS. For run-time bound and stability the smoke test uses gray radiation with aerosol and volcanic disabled (`--radiation gray --no-aerosol --no-volcanic`); it confirms the (grid, discretisation, forcing-loader) machinery wires up but does *not* validate the full forcing stack on every grid. The Voronoi/MPAS branch additionally requires `--turbulence none` (the TKE bridge expects cell-centred winds, MPAS stores edge-normal winds) and `--dt 60` (the MPAS hydrostatic dycore is unstable at the default 600 s step despite a CFL diagnostic of 0.09). Spectral and MPAS run paths bypass `DiagnosticCollector`, so detailed zonal/profile diagnostics are unavailable on those grids; production science runs target cubed-sphere or lat-lon.

16.4 Restart and Reproducibility

Each restart file includes a `.meta.json` sidecar with SHA-256 state digest, config hash, platform identifier, JAX version, x64 mode, and git hash.

Chapter 17

Verification and Testing

The legoESM codebase comprises approximately 385 source modules ($\sim 102,000$ lines of code), validated by 3,489 tests across 216 test files.

17.1 Williamson Standard Test Cases

The shallow water dynamical cores are verified against the standard Williamson test suite (Williamson et al., 1992), including:

- **Case 1:** Advection of a cosine bell;
- **Case 2:** Global steady-state nonlinear geostrophic flow;
- **Case 5:** Zonal flow over an isolated mountain;
- **Case 6:** Rossby–Haurwitz wave.

A comprehensive diagnostic suite (`williamson_diagnostic.py`) computes L_1 , L_2 , and L_∞ error norms against analytical solutions where available, and performs cross-discretisation comparison across all five grid families (cubed-sphere C-D, spectral, finite-volume lat-lon, C-grid lat-lon, and MPAS/Voronoi).

17.2 Spectral Dynamical Core Test Suite

The spectral transform dynamical cores (shallow water and primitive equation) are validated by a dedicated comprehensive test suite covering:

- **Conservation:** Total mass, energy, and enstrophy conservation to machine precision (shallow water) and within bounded drift (primitive equation with hyperdiffusion);
- **Stability:** Long-integration stability under Held–Suarez forcing at T21/L40 and T42/L40 resolutions;
- **Spectral convergence:** Verification that error norms decrease with increasing truncation wavenumber;
- **Semi-implicit correctness:** Validation that the implicit gravity-wave solver matches reference solutions and permits CFL numbers > 1 for gravity-wave modes.

17.3 Cross-Discretisation Intercomparison

All five shallow water discretisations are compared on the same test problems, verifying consistent convergence rates and inter-scheme agreement in error norms. This ensures that the shared physics

interface produces consistent results regardless of the dynamical core choice.

17.4 MPI Distributed Tests

Dedicated distributed test suites (`tests/distributed/`) verify:

- **Halo exchange correctness:** Scalar and vector 4D halo exchange on 2–6 ranks matches serial reference within round-off;
- **MPI differentiability:** `jax.grad` through halo-exchange and allreduce operations produces gradients matching single-rank reference (`test_mpi_differentiability.py`);
- **MPI driver integration:** Full model stepping under MPI produces results consistent with serial execution (`test_mpi_driver.py`).

Chapter 18

Physical Constants

Symbol	Name	Value	Unit
g	Gravitational acceleration	9.80616	m s^{-2}
Ω	Earth rotation rate	7.292×10^{-5}	rad s^{-1}
$R_{\oplus}$	Earth mean radius	6.371×10^6	m
R_d	Dry-air gas constant	287.05	$\text{J kg}^{-1} \text{K}^{-1}$
c_p	Dry-air specific heat (c_p)	1004.64	$\text{J kg}^{-1} \text{K}^{-1}$
c_v	Dry-air specific heat (c_v)	717.56	$\text{J kg}^{-1} \text{K}^{-1}$
κ	Poisson constant R_d/c_p	0.2857	—
R_v	Water-vapour gas constant	461.51	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pv}	Specific heat of water vapour	1846	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pw}	Specific heat of liquid water	4218	$\text{J kg}^{-1} \text{K}^{-1}$
c_{pi}	Specific heat of ice	2106	$\text{J kg}^{-1} \text{K}^{-1}$
L_v	Latent heat of vapourisation	2.501×10^6	J kg^{-1}
L_s	Latent heat of sublimation	2.834×10^6	J kg^{-1}
L_f	Latent heat of fusion	3.337×10^5	J kg^{-1}
ε	R_d/R_v	0.622	—
T_{freeze}	Freezing point of water	273.15	K
ρ_w	Liquid water density	1000	kg m^{-3}
ρ_i	Ice density	917	kg m^{-3}
p_0	Reference pressure	10^5	Pa
S_0	Total solar irradiance	1361	W m^{-2}
σ_{SB}	Stefan–Boltzmann constant	5.670×10^{-8}	$\text{W m}^{-2} \text{K}^{-4}$
κ_{vK}	von Kármán constant	0.4	—
ρ_0	Ocean reference density	1025	kg m^{-3}

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